

# 7.5 inch
# E-paper Display Series

## GDEY075T7-T01

Dalian Good Display Co., Ltd.

# Product Specifications




| | |
|---|---|
| **Customer** | **Standard** |
| **Description** | **7.5” E-PAPER DISPLAY** |
| **Model Name** | **GDEY075T7-T01** |
| **Date** | **2024/03/10** |
| **Revision** | **1.0** |

| | Design Engineering | | |
|---|---|---|---|
| | **Approval** | **Check** | **Design** |
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## 1.Over View

GDEY075T7-T01 is an Active Matrix Electrophoretic Display (AMEPD), with touch screen. The7.5” active area contains 800×480 pixels, and has 1-bit B/W full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

## 2.Features

- 800×480 pixels display
- High contrast
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Waveform stored in flash memory
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM,Gate and
- Source driving voltage
- I2C Signal Master Interface to read external temperature sensor/ built-in temperature sensor
- Available in COG package IC thickness 300um

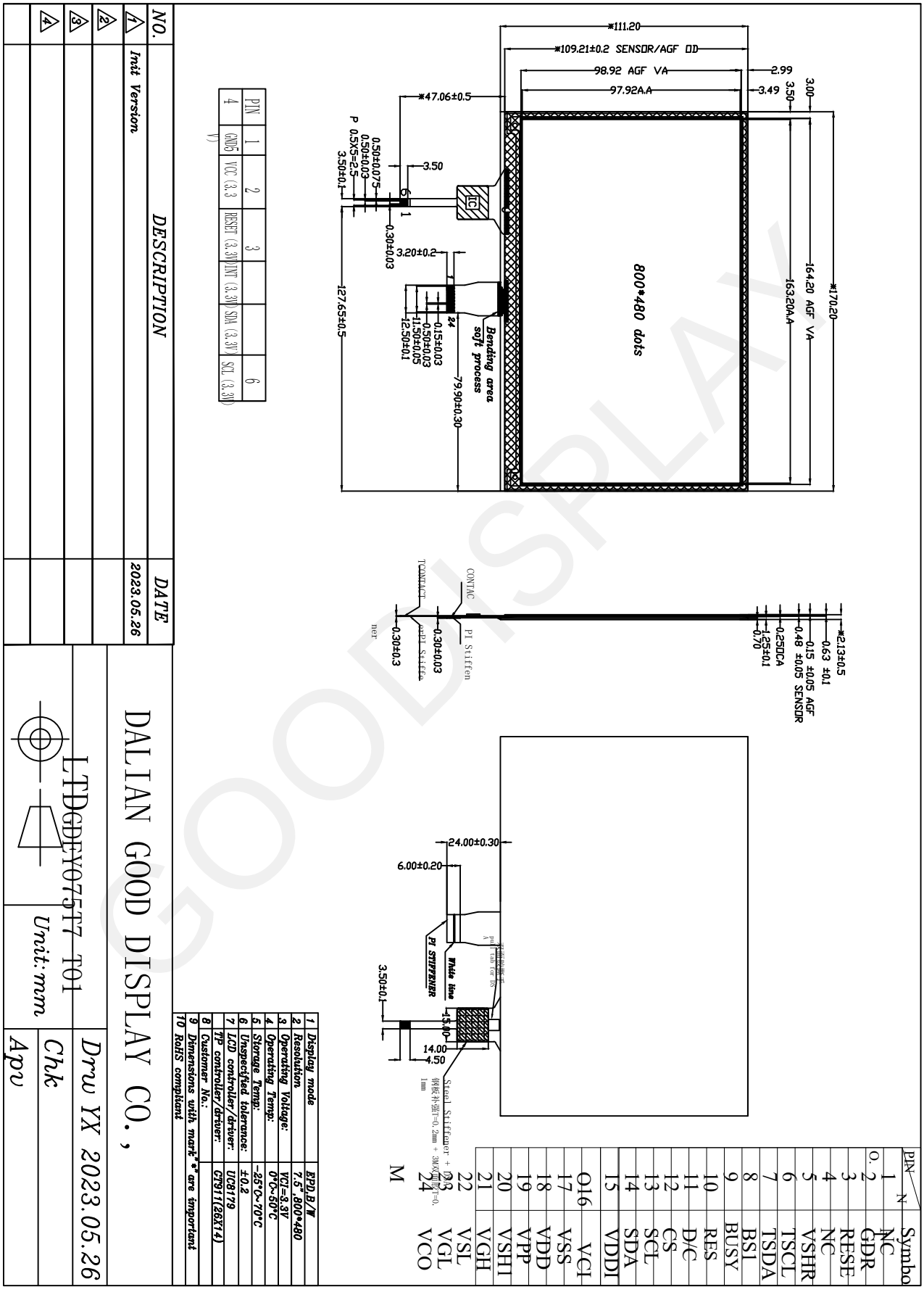
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## 3. Mechanical Specifications

| Parameter | Specifications | Unit | Remark |
|---|---|---|---|
| Screen Size | 7.5 | Inch | |
| Display Resolution | 800 (H)×480(V) | Pixel | Dpi:124 |
| Active Area | 163.20(H)×97.92(V) | mm | |
| Pixel Pitch | 0.204×0.204 | mm | |
| Pixel Configuration | Square | | |
| Outline Dimension | 170.20(H)×111.20 (V) ×2.13 (D) | mm | |
| Weight | 65±0.5 | g | |

# 4. Mechanical Drawing of EPD module

| CUSTOMER'S APPROVED: | DATE: | PAGE: |
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## 5. Input /Output Pin Assignment

| No. | Name | I/O | Description | Remark |
|---|---|---|---|---|
| 1 | NC | | Do not connect with other NC pins | Keep Open |
| 2 | GDR | O | N-Channel MOSFET Gate Drive Control | |
| 3 | RESE | I | Current Sense Input for the Control Loop | |
| 4 | NC | NC | Do not connect with other NC pins | Keep Open |
| 5 | VSHR | C | Positive Source driving voltage(Red) | |
| 6 | TSCL | O | $I^2C$ Interface to digital temperature sensor Clock pin | |
| 7 | TSDA | I/O | $I^2C$ Interface to digital temperature sensor Data pin | |
| 8 | BS1 | I | Bus Interface selection pin | Note 5-5 |
| 9 | BUSY | O | Busy state output pin | Note 5-4 |
| 10 | RES | I | Reset signal input. Active Low. | Note 5-3 |
| 11 | D/C | I | Data /Command control pin | Note 5-2 |
| 12 | CS | I | Chip select input pin | Note 5-1 |
| 13 | SCL | I | Serial Clock pin (SPI) | |
| 14 | SDA | I/O | Serial Data pin (SPI) | |
| 15 | VDDIO | P | Power Supply for interface logic pins It should be connected with VCI | |
| 16 | VCI | P | Power Supply for the chip | |
| 17 | VSS | P | Ground | |
| 18 | VDD | C | Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS | |
| 19 | VPP | P | FOR TEST | |
| 20 | VSH1 | C | Positive Source driving voltage | |
| 21 | VGH | C | Power Supply pin for Positive Gate driving voltage and VSH1 | |
| 22 | VSL | C | Negative Source driving voltage | |
| 23 | VGL | C | Power Supply pin for Negative Gate driving voltage VCOM and VSL | |
| 24 | VCOM | C | VCOM driving voltage | |

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I = Input Pin, O =Output Pin, I/O = Bi-directional Pin (Input/output), P = Power Pin, C =Capacitor Pin

**Note 5-1:** This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CSB is pulled Low.

**Note 5-2:** This pin (DC) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled Low, the data will be interpreted as command.

**Note 5-3:** This pin (RST) is reset signal input. The Reset is active Low

**Note 5-4:** This pin (BUSY) is BUSY state output pin. When BUSY is low, the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put BUSY pin low when the driver IC is working such as:

Outputting display waveform; or

Programming with OTP

Communicating with digital temperature sensor

| BS1 State | MCU Interface |
|---|---|
| L | 4-lines serial peripheral interface(SPI) |
| H | 3- lines serial peripheral interface(SPI) - 9 bits SPI |

**Table: Bus interface selection**

# 6.Electrical Characteristics

## 6.1 Absolute Maximum Rating

| Parameter | Symbol | Rating | Unit |
|---|---|---|---|
| Logic supply voltage | VCI | -0.3 to +6.0 | V |
| Logic Input voltage | VIN | -0.3 to VCI +0.3 | V |
| Operating Temp range | TOPR | 0 to +50 | °C |
| Storage Temp range | TSTG | -25 to+70 | °C |
| Optimal Storage Humidity | HSTGo | 55±10 | %RH |
| Touch Panel Operating Voltage | - | 2.8~3.6 | V |
| Touch Panel Operating Current | - | 4.32 | mA |
| Touch Panel Standby Current | - | 55 | μA |

**Note:**

Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables

## 6.2 DC Characteristics

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23ºC

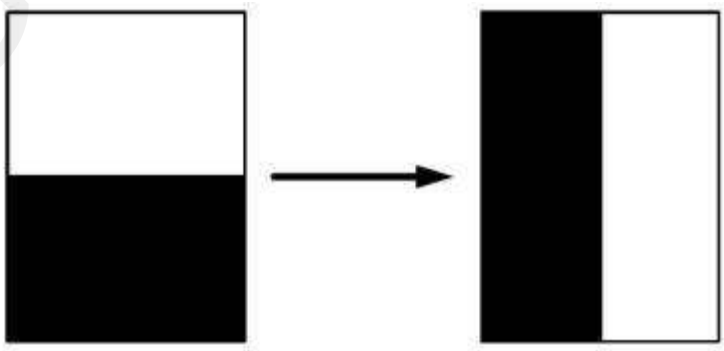
| Parameter | Symbol | Conditions | Applicable pin | Min. | Typ. | Max | Units |
|---|---|---|---|---|---|---|---|
| Single ground | $V_{SS}$ | - | | - | 0 | - | V |
| Logic supply voltage | $V_{CI}$ | - | VCI | 2.5 | 3.3 | 3.6 | V |
| Core logic voltage | $V_{DD}$ | | VDD | 2.5 | 3.3 | 3.6 | V |
| High level input voltage | $V_{IH}$ | - | - | 0.7 $V_{CI}$ | - | - | V |
| Low level input voltage | $V_{IL}$ | - | - | - | - | 0.3 $V_{CI}$ | V |
| High level output voltage | $V_{OH}$ | IOH = -400uA | - | VCI-0.4 | - | - | V |
| Low level output voltage | $V_{OL}$ | IOL = 400uA | - | 0 | - | 0.4 | V |
| Typical operating current | Iopr_$V_{CI}$ | $V_{CI}$ =3.3V | - | - | 8 | 12 | mA |
| Full update time | - | 25 ºC | - | - | 3 | - | sec |
| Partial update time | - | 25 ºC | - | - | 0.3 | - | sec |
| Fast update time | - | 25 ºC | - | - | 1.5 | - | sec |
| Sleep mode current | Islp_$V_{CI}$ | DC/DC off<br>No clock<br>No input load<br>Ram data retain | - | - | 25 | | uA |
| Deep sleep mode current | Idslp_$V_{CI}$ | DC/DC off<br>No clock<br>No input load<br>Ram data not retain | - | - | 2 | 5 | uA |

**Notes:**

1. The typical power is measured with following transition from horizontal 2 scale pattern to vertical 2 scale pattern.

2.The deep sleep power is the consumed power when the panel controller is in deep sleep mode.

3.The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by GOOD DISPLAY



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## 6.3 AC Characteristics

### 6.3.1 MCU Interface Selection

Provides 3-wire/4-wire serial interface for command and display data transferred from the MCU. The serial interface supports 8-bit mode. Data can be input/output by clocks while the chip is active (CSB =LOW). While input, data are written in order from MSB at the clock rising edge. When too many parameters are input, the chip accepts only defined parameters, and ignores undefined ones.

| BS | Interface | CSB | DC | SCL | SDA |
|---|---|---|---|---|---|
| High | 3-wire SPI | Available | Fix to GND | Available | Available |
| Low | 4-wire SPI | Available | Available | Available | Available |

### 6.3.2 3 wire SPI format.

Data / Command is recognized with the first bit transferred. Data are transferred in the unit of 9 bits. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 9 bits. (The serial counter is reset at the rising edge of the CSB signal.)

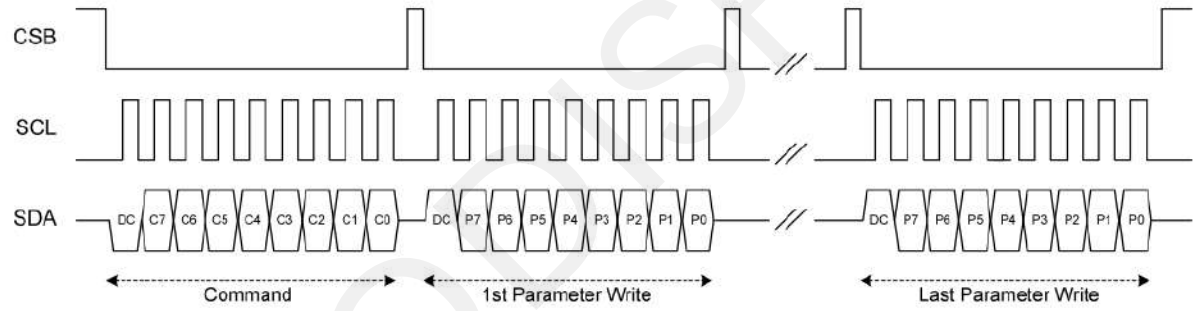



**Figure:** 3-wire SPI write operation

The MSB bit of data will be output at SDA pin after the 1st SCL falling edge, if the 1st input data at SDA is high.

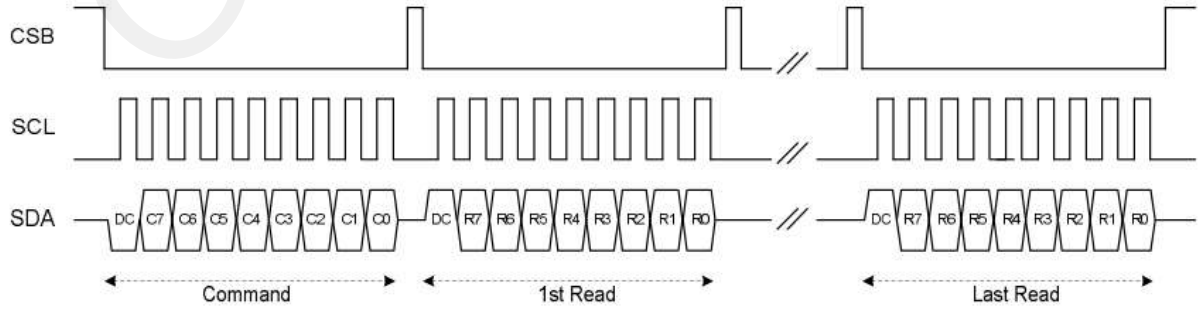



**Figure:** 3-wire SPI read operation

### 6.3.3 4 wire SPI format

Data / Command is recognized with DC pin. Data are transferred in the unit of 8 bits. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 8 bits. (The serial counter is reset at the rising edge of the CSB signal.)

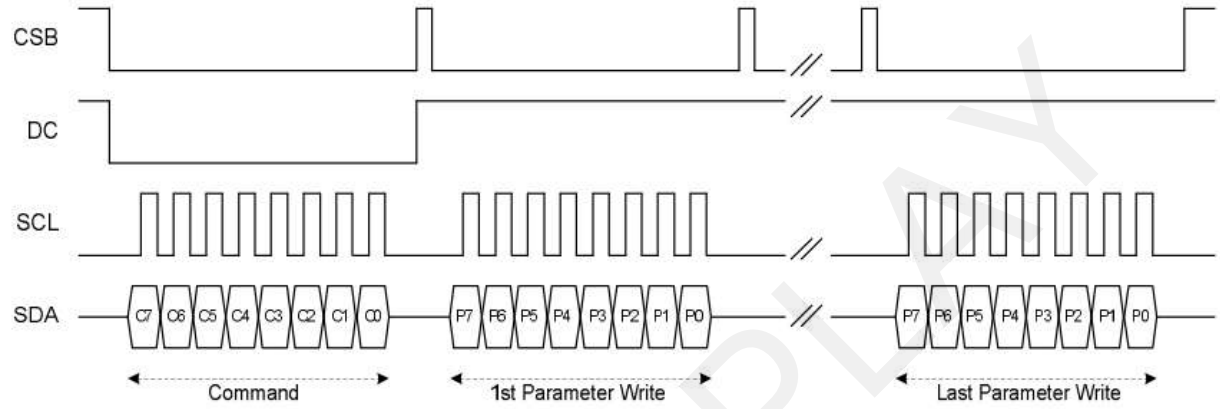



**Figure:** 4-wire SPI write operation

The MSB bit of data will be output at SDA pin after the CSB falling edge, if DC pin is High.

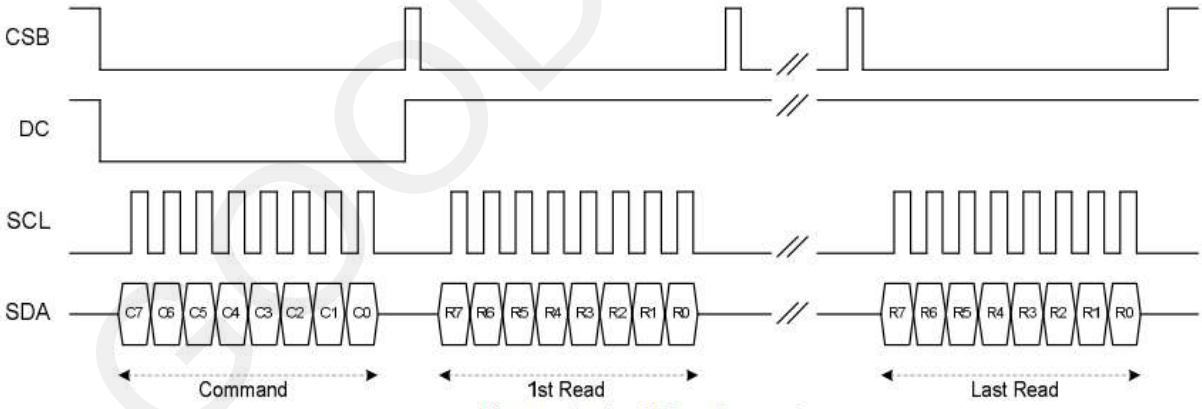



**Figure:** 4-wire SPI read operation

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### 6.3.4 3 wire dual SPI format.

Data / Command is recognized with the first bit transferred at SDA. Data are transferred in the unit of 5 SPI clocks. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 5 SPI clocks. (The serial counter is reset at the rising edge of the CSB signal.) In 3-wire dual SPI mode, SDA and SDA1 are only input mode for data write transmission.

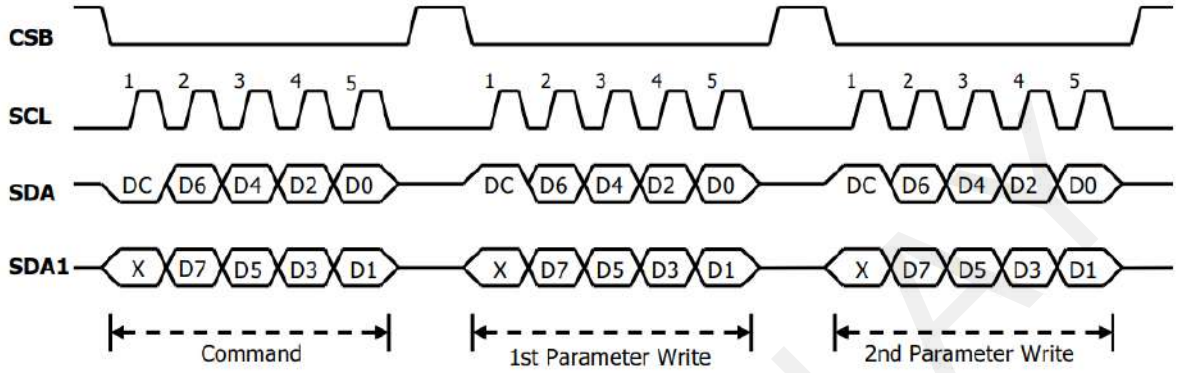



**Figure:** 3-wire dual SPI write operation

### 6.3.5 4 wire dual SPI format.

Data / Command is recognized with DC pin. Data are transferred in the unit of 4 SPI clocks. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 4 SPI clocks. (The serial counter is reset at the rising edge of the CSB signal.) In 4-wire dual SPI mode, SDA and SDA1 are only input mode for data write transmission.

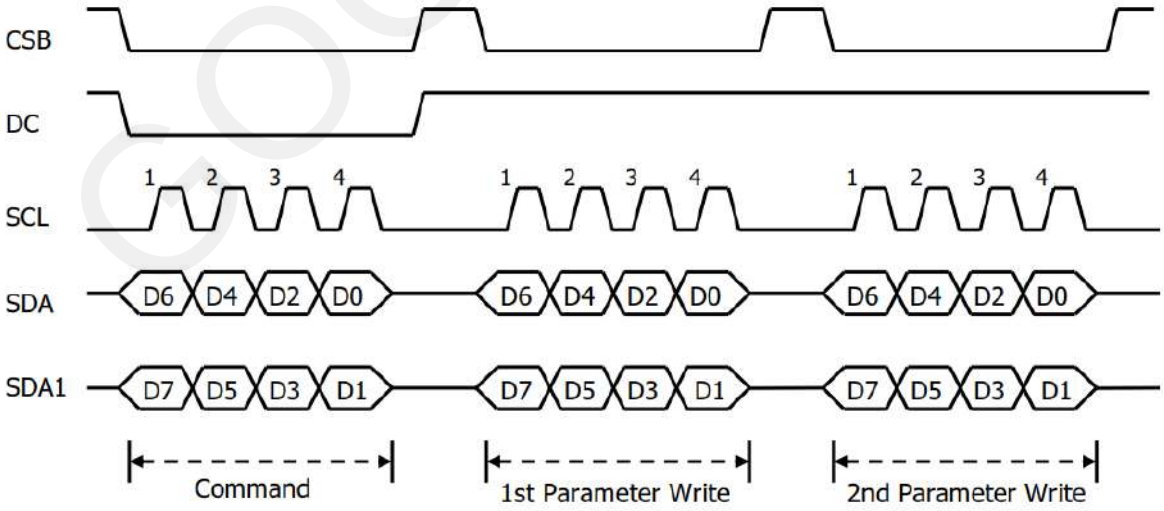



**Figure:** 4-wire dual SPI write operation

## 6.4 Timing Characteristics.

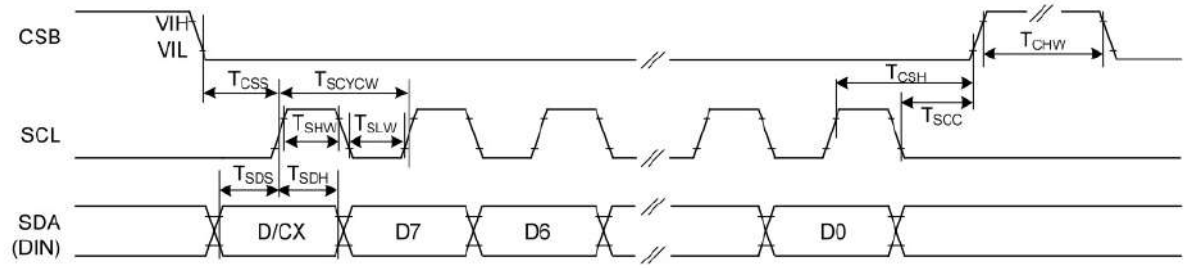



**Figure:** 3-wire Serial Interface Characteristics (Write mode)

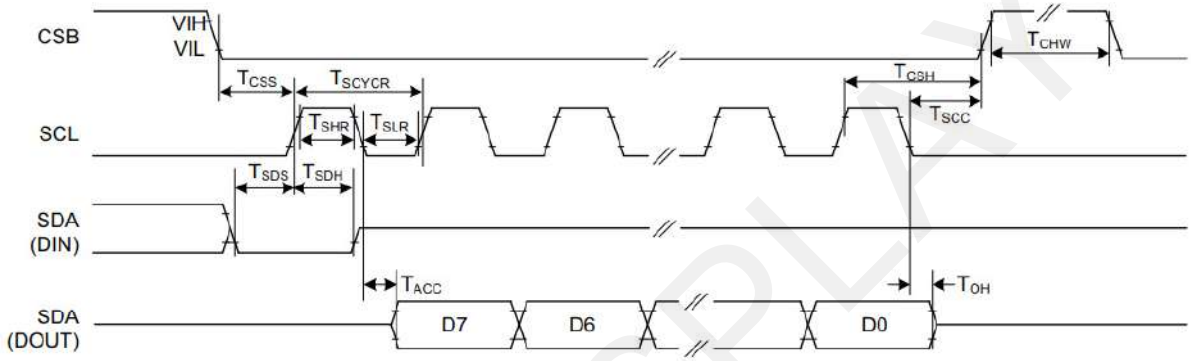



**Figure:** 3-wire Serial Interface Characteristics (Read mode)

| Symbol | Signal / Parameter | Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $T_{CSS}$ | CSB | Chip select setup time | 60 | | | ns |
| $T_{CSH}$ | | Chip select hold time | 65 | | | ns |
| $T_{SCC}$ | | Chip select setup time | 20 | | | ns |
| $T_{CHW}$ | | Chip select setup time | 40 | | | ns |
| $T_{SCYCW}$ | SCL | Serial clock cycle (Write) | 100 | | | ns |
| $T_{SHW}$ | | SCL "H" pulse width (Write) | 35 | | | ns |
| $T_{SLW}$ | | SCL "L" pulse width (Write) | 35 | | | ns |
| $T_{SCYCR}$ | | Serial clock cycle (Read) | 230 | | | ns |
| $T_{SHR}$ | | SCL "H" pulse width (Read) | 60 | | | ns |
| $T_{SLR}$ | | SCL "L" pulse width (Read) | 60 | | | ns |
| $T_{SDS}$ | SDA (DIN) | Data setup time | 30 | | | ns |
| $T_{SDH}$ | | Data hold time | 30 | | | ns |
| $T_{ACC}$ | SDA (DOUT) | Access time | | | 230 | ns |
| $T_{OH}$ | | Output disable time | 15 | | | ns |

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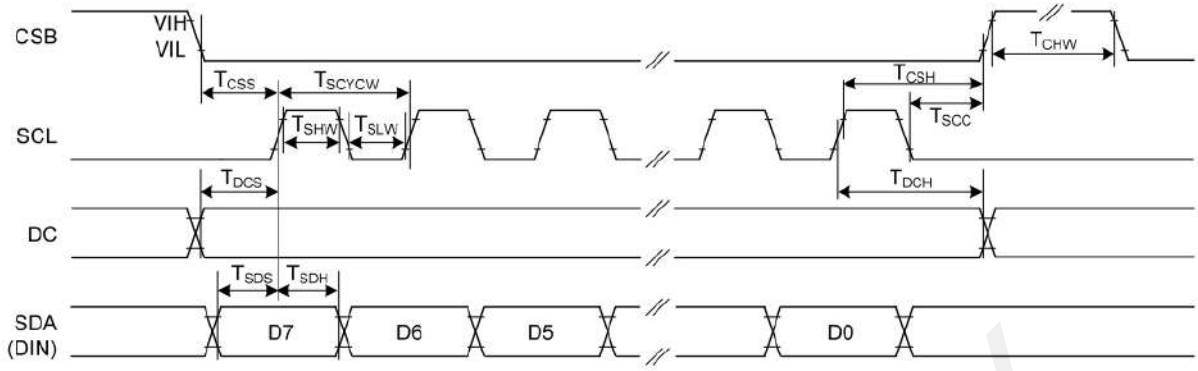



**Figure:** 4-wire Serial Interface Characteristics (Write mode)

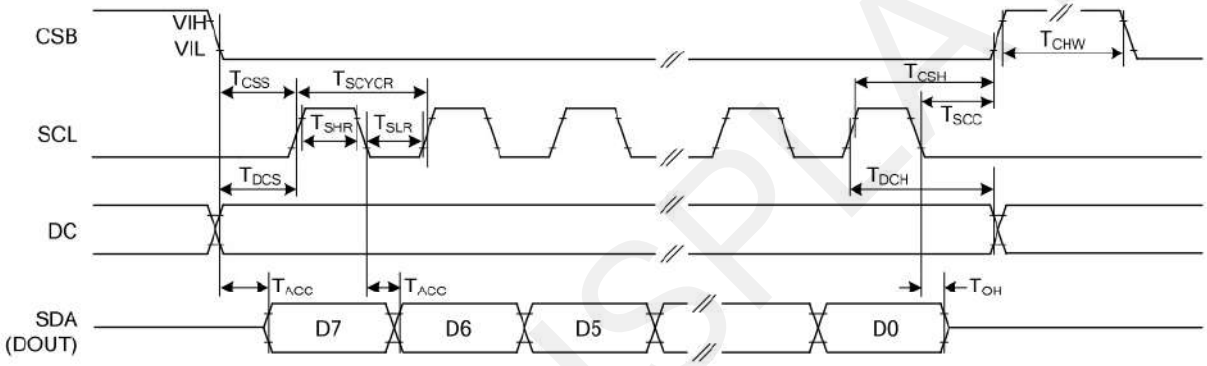



**Figure:** 4-wire Serial Interface Characteristics (Read mode)

| Symbol | Signal / Parameter | Conditions | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $T_{CSS}$ | CSB | Chip select setup time | 60 | | | ns |
| $T_{CSH}$ | | Chip select hold time | 65 | | | ns |
| $T_{SCC}$ | | Chip select setup time | 20 | | | ns |
| $T_{CHW}$ | | Chip select setup time | 40 | | | ns |
| $T_{SCYCW}$ | SCL | Serial clock cycle (Write) | 100 | | | ns |
| $T_{SHW}$ | | SCL "H" pulse width (Write) | 35 | | | ns |
| $T_{SLW}$ | | SCL "L" pulse width (Write) | 35 | | | ns |
| $T_{SCYCR}$ | | Serial clock cycle (Read) | 230 | | | ns |
| $T_{SHR}$ | | SCL "H" pulse width (Read) | 60 | | | ns |
| $T_{SLR}$ | | SCL "L" pulse width (Read) | 60 | | | ns |
| $T_{DCS}$ | DC | DC setup time | 30 | | | ns |
| $T_{DCH}$ | | DC hold time | 30 | | | ns |
| $T_{SDS}$ | SDA (DIN) | Data setup time | 30 | | | ns |
| $T_{SDH}$ | | Data hold time | 30 | | | ns |
| $T_{ACC}$ | SDA (DOUT) | Access time | | | 230 | ns |
| $T_{OH}$ | | Output disable time | 15 | | | ns |

# 7. Command Table

W/R: 0: Write Cycle 1: Read Cycle C/D: 0: Command / 1: Data D7~D0: –: Don't Care #: Valid Data

| # | Command | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Registers | Default |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Panel Setting (PSR) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | 00H |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | REG, KW/R, UD, SHL, SHD_N, RST_N | 0FH |
| 2 | Power Setting (PWR) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | | 01H |
| | | 0 | 1 | -- | -- | -- | # | -- | # | # | # | BD_EN, VSR_EN, VS_EN, VG_EN | 07H |
| | | 0 | 1 | -- | -- | -- | # | -- | # | # | # | VCOM_SLEW, VG_LVL[2:0] | 17H |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | VDH_LVL[5:0] | 3AH |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | VDL_LVL[5:0] | 3AH |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | VDHR_LVL[5:0] | 03H |
| 3 | Power OFF (POF) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | | 02H |
| 4 | Power OFF Sequence Setting (PFS) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | | 03H |
| | | 0 | 1 | -- | -- | # | # | -- | -- | -- | -- | T_VDS_OFF[1:0] | 00H |
| 5 | Power ON (PON) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | | 04H |
| 6 | Power ON Measure (PMES) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | | 05H |
| 7 | Booster Soft Start (BTST) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | | 06H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | BT_PHA[7:0] | 17H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | BT_PHB[7:0] | 17H |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | BT_PHC1[5:0] | 17H |
| | | 0 | 1 | # | -- | # | # | # | # | # | # | PHC2_EN, BT_PHC2[5:0] | 17H |
| 8 | Deep sleep (DSLP) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | | 07H |
| | | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | Check code | A5H |
| 9 | Display Start Transmission 1 (DTM1, White/Black Data) (x-byte command) | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | K/W or OLD Pixel Data (800x600): | 10H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | KPXL[1:8] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | : | : |
| | | 0 | 1 | # | # | # | # | # | # | # | # | KPXL[n-7:n] | - |
| 10 | Data Stop (DSP) | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | | 11H |
| | | 1 | 1 | # | -- | -- | -- | -- | -- | -- | -- | | 00H |
| 11 | Display Refresh (DRF) | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | | 12H |
| 12 | Display Start transmission 2 (DTM2, Red Data) (x-byte command) | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | Red or NEW Pixel Data (800x600): | 13H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | RPXL[1:8] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | : | : |
| | | 0 | 1 | # | # | # | # | # | # | # | # | RPXL[n-7:n] | - |
| 13 | Dual SPI | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | | 15H |
| | | 1 | 1 | -- | -- | # | # | -- | -- | -- | -- | MM_EN, DUSPI_EN | 00H |
| 14 | Auto Sequence (AUTO) | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | | 17H |
| | | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | Check code | A5H |
| 15 | VCOM LUT (LUTC) (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | | 20H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |
| 16 | W2W LUT (LUTWW) (43-byte command, structure of bytes 2~7 repeated 7 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | | 21H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |

| # | Command | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Registers | Default |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 17 | K2W LUT (LUTKW / LUTR) (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | | 22H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |
| 18 | W2K LUT (LUTWK / LUTW) (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | | 23H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |
| 19 | K2K LUT (LUTKK / LUTK) (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | | 24H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |
| 20 | Border LUT (43-byte command, structure of bytes 2~7 repeated 7 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | | 25H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Level select-0~3[1:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-0[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-1[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-2[7:0] | - |
| | | 0 | 1 | : | : | : | : | : | : | : | : | Number of frames-3[7:0] | - |
| | | 0 | 1 | # | # | # | # | # | # | # | # | Times to repeat[7:0] | - |
| 21 | LUT option (LUTOPT) | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | | 2AH |
| | | 0 | 1 | # | # | -- | -- | -- | -- | -- | -- | STATE_XON[9:8] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | STATE_XON[7:0] | 00H |
| 22 | KW LUT option (KWOPT) | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | | 2BH |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | ATRED, NORED | 00H |
| | | 0 | 1 | # | # | -- | -- | -- | -- | -- | -- | KWE[9:8] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | KWE[7:0] | 00H |
| 23 | PLL control (PLL) | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | | 30H |
| | | 0 | 1 | -- | -- | -- | -- | # | # | # | # | FRS[3:0] | 06H |
| 24 | Temperature Sensor Calibration (TSC) | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | | 40H |
| | | 1 | 1 | # | # | # | # | # | # | # | # | D[10:3] / TS[7:0] | 00H |
| | | 1 | 1 | # | # | # | -- | -- | -- | -- | -- | D[2:0] / - | 00H |
| 25 | Temperature Sensor Selection (TSE) | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | | 41H |
| | | 0 | 1 | # | -- | -- | -- | # | # | # | # | TSE,TO[3:0] | 00H |
| 26 | Temperature Sensor Write (TSW) | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | | 42H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | WATTR[7:0] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | WMSB[7:0] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | WLSB[7:0] | 00H |
| 27 | Temperature Sensor Read (TSR) | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | | 43H |
| | | 1 | 1 | # | # | # | # | # | # | # | # | RMSB[7:0] | 00H |
| | | 1 | 1 | # | # | # | # | # | # | # | # | RLSB[7:0] | 00H |
| 28 | Panel Break Check (PBC) | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | | 44H |
| | | 1 | 1 | -- | -- | -- | -- | -- | -- | -- | # | PSTA | 00H |
| 29 | VCOM and data interval setting (CDI) | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | | 50H |
| | | 0 | 1 | # | -- | # | # | -- | -- | # | # | BDZ, BDV[1:0], DDX[1:0] | 31H |
| | | 0 | 1 | -- | -- | -- | -- | # | # | # | # | CDI[3:0] | 07H |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 31 | End Voltage Setting (EVS) | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | | 52H |
| | | 0 | 1 | -- | -- | -- | -- | # | -- | # | # | VCEND, BDEND[1:0] | 02H |
| 32 | TCON setting (TCON) | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | | 60H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | S2G[3:0], G2S[3:0] | 22H |
| 33 | Resolution setting (TRES) | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | | 61H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | HRES[9:8] | 03H |
| | | 0 | 1 | # | # | # | # | # | 0 | 0 | 0 | HRES[7:3] | 20H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | VRES[9:0] | 02H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | | 58H |
| 34 | Gate/Source Start setting (GSST) | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | | 65H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | HST[9:8] | 00H |
| | | 0 | 1 | # | # | # | # | # | 0 | 0 | 0 | HST[7:3] | 00H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | VST[9:0] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | | 00H |
| 35 | Revision (REV) | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | | 70H |
| | | 1 | 1 | # | # | # | # | # | # | # | # | PROD_REV[23:16] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | PROD_REV[15:8] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | PROD_REV[7:0] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | LUT_REV[23:16] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | LUT_REV[15:8] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | LUT_REV[7:0] | FFH |
| | | 1 | 1 | # | # | # | # | # | # | # | # | CHIP_REV[7:0] | 0CH |
| 36 | Get Status (FLG) | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | | 71H |
| | | 1 | 1 | -- | # | # | # | # | # | # | # | PTL_FLAG ,I²C_ERR, I²C_BUSYN, DATA_FLAG, PON, POF, BUSY_N | 13H |
| 37 | Auto Measurement VCOM (AMV) | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | 80H |
| | | 0 | 1 | -- | -- | # | # | # | # | # | # | AMVT[1:0], XON,AMVS, AMV, AMVE | 10H |
| 38 | Read VCOM Value (VV) | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | | 81H |
| | | 1 | 1 | -- | # | # | # | # | # | # | # | VV[6:0] | 00H |
| 39 | VCOM_DC Setting (VDCS) | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | | 82H |
| | | 0 | 1 | -- | # | # | # | # | # | # | # | VDCS[6:0] | 00H |
| 40 | Partial Window (PTL) | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | | 90H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | HRST[9:8] | 00H |
| | | 0 | 1 | # | # | # | # | # | 0 | 0 | 0 | HRST[7:3] | 00H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | HRED[9:8] | 03H |
| | | 0 | 1 | # | # | # | # | # | 1 | 1 | 1 | HRED[7:3] | 1FH |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | VRST[9:0] | 00H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | | 00H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | VRED[8:0] | 02H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | | 57H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | -- | # | PT_SCAN | 01H |
| 41 | Partial In (PTIN) | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | | 91H |
| 42 | Partial Out (PTOUT) | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | | 92H |
| 43 | Program Mode (PGM) | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | | A0H |
| 44 | Active Programming (APG) | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | | A1H |
| 45 | Read OTP (ROTP) | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | | A2H |
| | | 1 | 1 | # | # | # | # | # | # | # | # | Data of Address = 000h | N/A |
| | | 1 | 1 | : | : | : | : | : | : | : | : | : | N/A |
| | | 1 | 1 | # | # | # | # | # | # | # | # | Data of Address = n | N/A |
| 46 | Cascade Setting (CCSET) | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | | E0H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | TSFIX, CCEN | 00H |
| 47 | Power Saving (PWS) | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | | E3H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | VCOM_W[3:0], SD_W[3:0] | 00H |
| 48 | LVD Voltage Select (LVSEL) | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | | E4H |
| | | 0 | 1 | -- | -- | -- | -- | -- | -- | # | # | LVD_SEL[1:0] | 03H |
| 49 | Force Temperature (TSSET) | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | | E5H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | TS_SET[7:0] | 00H |
| 50 | Temperature Boundary Phase-C2 (TSBDRY) | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | | E7H |
| | | 0 | 1 | # | # | # | # | # | # | # | # | TSBDRY_PHC2[7:0] | 00H |

## COMMAND DESCRIPTION

W/R: 0: Write Cycle / 1: Read Cycle C/D: 0: Command / 1: Data D7-D0: –: Don't Care

### (1) PANEL SETTING (PSR) (REGISTER: R00H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Setting the panel | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 00H |
| | 0 | 1 | - | - | REG | KW/R | UD | SHL | SHD_N | RST_N | 0FH |

**REG:** LUT selection

**0: LUT from OTP. (Default)**
1: LUT from register.

**KW/R:** Black / White / Red

**0: Pixel with Black/White/Red, KWR mode. (Default)**
1: Pixel with Black/White, KW mode.

**UD:** Gate Scan Direction

0: Scan down. First line to Last line: Gn-1 → Gn-2 → Gn-3 → … → G0
**1: Scan up. (Default)** First line to Last line: G0 → G1 → G2 → … … . → Gn-1

**SHL:** Source Shift Direction

0: Shift left. First data to Last data: Sn-1 → Sn-2 → Sn-3 → … → S0
**1: Shift right. (Default)** First data to Last data: S0 → S1 → S2 → … … . → Sn-1

**SHD_N:** Booster Switch

0: Booster OFF
**1: Booster ON (Default)**

When SHD_N becomes LOW, charge pump will be turned OFF, register and SRAM data will keep until VDD OFF. And Source/Gate/Border/VCOM will be released to floating.

**RST_N:** Soft Reset

0: Reset. Booster OFF, Register data are set to their default values, all drivers will be reset, and all functions will be disabled. Source/Gate/Border/VCOM will be released to floating.

**1: No effect (Default).**

### (2) POWER SETTING (PWR) (R01H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Selecting Internal/External Power | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 01H |
| | 0 | 1 | - | - | - | BD_EN | - | VSR_EN | VS_EN | VG_EN | 07H |
| | 0 | 1 | - | - | - | VCOM _SLEW | - | VG_LVL[2:0] | | | 17H |
| | 0 | 1 | - | - | VDH_LVL[5:0] | | | | | | 3AH |
| | 0 | 1 | - | - | VDL_LVL[5:0] | | | | | | 3AH |
| | 0 | 1 | - | - | VDHR_LVL[5:0] | | | | | | 03H |

**BD_EN:** Border LDO enable
**0 : Border LDO disable (Default)**
Border level selection: 00b: VCOM 01b: VDH 10b: VDL 11b: VDHR
1 : Border LDO enable
Border level selection: 00b: VCOM 01b: VBH(VCOM-VDL) 10b:VBL(VCOM-VDH) 11b: VDHR

**VSR_EN:** Source LV power selection
0 : External source power from VDHR pins
**1 : Internal DC/DC function for generating VDHR. (Default)**

**VS_EN:** Source power selection
0 : External source power from VDH/VDL pins
**1 : Internal DC/DC function for generating VDH/VDL. (Default)**

**VG_EN:** Gate power selection
0 : External gate power from VGH/VGL pins
**1 : Internal DC/DC function for generating VGH/VGL. (Default)**

**VCOM_SLEW:** VCOM slew rate selection for voltage transition. The value is fixed at "**1**".

**VG_LVL[2:0]:**VGH / VGL Voltage Level selection.

| VG_LVL[2:0] | VGH/VGL Voltage Level |
|---|---|
| 000 | VGH=9V, VGL= -9V |
| 001 | VGH=10V, VGL= -10V |
| 010 | VGH=11V, VGL= -11V |
| 011 | VGH=12V, VGL= -12V |
| 100 | VGH=17V, VGL= -17V |
| 101 | VGH=18V, VGL= -18V |
| 110 | VGH=19V, VGL= -19V |
| 111 **(Default)** | VGH=20V, VGL= -20V |

**VDH_LVL[5:0]:** Internal VDH power selection for K/W pixel.(**Default value: 111010b**)

| VDH_LVL | Voltage | VDH_LVL | Voltage | VDH_LVL | Voltage | VDH_LVL | Voltage |
|---|---|---|---|---|---|---|---|
| 000000 | 2.4 V | 010001 | 5.8 V | 100010 | 9.2 V | 110011 | 12.6 V |
| 000001 | 2.6 V | 010010 | 6.0 V | 100011 | 9.4 V | 110100 | 12.8 V |
| 000010 | 2.8 V | 010011 | 6.2 V | 100100 | 9.6 V | 110101 | 13.0 V |
| 000011 | 3.0 V | 010100 | 6.4 V | 100101 | 9.8 V | 110110 | 13.2 V |
| 000100 | 3.2 V | 010101 | 6.6 V | 100110 | 10.0 V | 110111 | 13.4 V |
| 000101 | 3.4 V | 010110 | 6.8 V | 100111 | 10.2 V | 111000 | 13.6 V |
| 000110 | 3.6 V | 010111 | 7.0 V | 101000 | 10.4 V | 111001 | 13.8 V |
| 000111 | 3.8 V | 011000 | 7.2 V | 101001 | 10.6 V | **111010** | **14.0 V** |
| 001000 | 4.0 V | 011001 | 7.4 V | 101010 | 10.8 V | 111011 | 14.2 V |
| 001001 | 4.2 V | 011010 | 7.6 V | 101011 | 11.0 V | 111100 | 14.4 V |
| 001010 | 4.4 V | 011011 | 7.8 V | 101100 | 11.2 V | 111101 | 14.6 V |
| 001011 | 4.6 V | 011100 | 8.0 V | 101101 | 11.4 V | 111110 | 14.8 V |
| 001100 | 4.8 V | 011101 | 8.2 V | 101110 | 11.6 V | 111111 | 15.0 V |
| 001101 | 5.0 V | 011110 | 8.4 V | 101111 | 11.8 V | | |
| 001110 | 5.2 V | 011111 | 8.6 V | 110000 | 12.0 V | | |
| 001111 | 5.4 V | 100000 | 8.8 V | 110001 | 12.2 V | | |
| 010000 | 5.6 V | 100001 | 9.0 V | 110010 | 12.4 V | | |

**VDL_LVL[5:0]:** Internal VDL power selection for K/W pixel. (**Default value: 111010b**)

| VDL_LVL | Voltage | VDL_LVL | Voltage | VDL_LVL | Voltage | VDL_LVL | Voltage |
|---|---|---|---|---|---|---|---|
| 000000 | -2.4 V | 010001 | -5.8 V | 100010 | -9.2 V | 110011 | -12.6 V |
| 000001 | -2.6 V | 010010 | -6.0 V | 100011 | -9.4 V | 110100 | -12.8 V |
| 000010 | -2.8 V | 010011 | -6.2 V | 100100 | -9.6 V | 110101 | -13.0 V |
| 000011 | -3.0 V | 010100 | -6.4 V | 100101 | -9.8 V | 110110 | -13.2 V |
| 000100 | -3.2 V | 010101 | -6.6 V | 100110 | -10.0 V | 110111 | -13.4 V |
| 000101 | -3.4 V | 010110 | -6.8 V | 100111 | -10.2 V | 111000 | -13.6 V |
| 000110 | -3.6 V | 010111 | -7.0 V | 101000 | -10.4 V | 111001 | -13.8 V |
| 000111 | -3.8 V | 011000 | -7.2 V | 101001 | -10.6 V | **111010** | **-14.0 V** |
| 001000 | -4.0 V | 011001 | -7.4 V | 101010 | -10.8 V | 111011 | -14.2 V |
| 001001 | -4.2 V | 011010 | -7.6 V | 101011 | -11.0 V | 111100 | -14.4 V |
| 001010 | -4.4 V | 011011 | -7.8 V | 101100 | -11.2 V | 111101 | -14.6 V |
| 001011 | -4.6 V | 011100 | -8.0 V | 101101 | -11.4 V | 111110 | -14.8 V |
| 001100 | -4.8 V | 011101 | -8.2 V | 101110 | -11.6 V | 111111 | -15.0 V |
| 001101 | -5.0 V | 011110 | -8.4 V | 101111 | -11.8 V | | |
| 001110 | -5.2 V | 011111 | -8.6 V | 110000 | -12.0 V | | |
| 001111 | -5.4 V | 100000 | -8.8 V | 110001 | -12.2 V | | |
| 010000 | -5.6 V | 100001 | -9.0 V | 110010 | -12.4 V | | |

**VDHR_LVL[5:0]:** Internal VDHR power selection for Red pixel. (**Default value: 000011b**)

| VDHR_LVL | Voltage | VDHR_LVL | Voltage | VDHR_LVL | Voltage | VDHR_LVL | Voltage |
|---|---|---|---|---|---|---|---|
| 000000 | 2.4 V | 010001 | 5.8 V | 100010 | 9.2 V | 110011 | 12.6 V |
| 000001 | 2.6 V | 010010 | 6.0 V | 100011 | 9.4 V | 110100 | 12.8 V |
| 000010 | 2.8 V | 010011 | 6.2 V | 100100 | 9.6 V | 110101 | 13.0 V |
| **000011** | **3.0 V** | 010100 | 6.4 V | 100101 | 9.8 V | 110110 | 13.2 V |
| 000100 | 3.2 V | 010101 | 6.6 V | 100110 | 10.0 V | 110111 | 13.4 V |
| 000101 | 3.4 V | 010110 | 6.8 V | 100111 | 10.2 V | 111000 | 13.6 V |
| 000110 | 3.6 V | 010111 | 7.0 V | 101000 | 10.4 V | 111001 | 13.8 V |
| 000111 | 3.8 V | 011000 | 7.2 V | 101001 | 10.6 V | 111010 | 14.0 V |
| 001000 | 4.0 V | 011001 | 7.4 V | 101010 | 10.8 V | 111011 | 14.2 V |
| 001001 | 4.2 V | 011010 | 7.6 V | 101011 | 11.0 V | 111100 | 14.4 V |
| 001010 | 4.4 V | 011011 | 7.8 V | 101100 | 11.2 V | 111101 | 14.6 V |
| 001011 | 4.6 V | 011100 | 8.0 V | 101101 | 11.4 V | 111110 | 14.8 V |
| 001100 | 4.8 V | 011101 | 8.2 V | 101110 | 11.6 V | 111111 | 15.0 V |
| 001101 | 5.0 V | 011110 | 8.4 V | 101111 | 11.8 V | | |
| 001110 | 5.2 V | 011111 | 8.6 V | 110000 | 12.0 V | | |
| 001111 | 5.4 V | 100000 | 8.8 V | 110001 | 12.2 V | | |
| 010000 | 5.6 V | 100001 | 9.0 V | 110010 | 12.4 V | | |

### (3) POWER OFF (POF) (R02H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Turning OFF the power | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 02H |

After the Power OFF command, the driver will be powered OFF. Refer to the POWER MANAGEMENT section for the sequence.

This command will turn off booster, controller, source driver, gate driver, VCOM, and temperature sensor, but register data will be kept until VDD turned OFF or Deep Sleep Mode. Source/Gate/Border/VCOM will be released to floating.

### (4) POWER OFF SEQUENCE SETTING (PFS) (R03H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Setting Power OFF sequence | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 03H |
| | 0 | 1 | - | - | T_VDS_OFF[1:0] | | - | - | - | - | 00H |

**T_VDS_OFF[1:0]:** Source to gate power off interval time.

**00b: 1 frame (Default)** 01b: 2 frames 10b: 3 frames 11b: 4 frame

### (5) POWER ON (PON) (REGISTER: R04H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Turning ON the power | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 04H |

After the Power ON command, the driver will be powered ON. Refer to the POWER MANAGEMENT section for the sequence.

This command will turn on booster, controller, regulators, and temperature sensor will be activated for one-time sensing before enabling booster. When all voltages are ready, the BUSY_N signal will return to high.

### (6) POWER ON MEASURE (PMES) (R05H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Internal Bandgap Set | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 05H |

This command enables the internal bandgap, which will be cleared by the next POF.

### (7) BOOSTER SOFT START (BTST) (R06H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Booster Software Start Set | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 06H |
| | 0 | 1 | BT_PHA[7:6] | | BT_PHA[5:3] | | | BT_PHA[2:0] | | | 17H |
| | 0 | 1 | BT_PHB[7:6] | | BT_PHB[5:3] | | | BT_PHB[2:0] | | | 17H |
| | 0 | 1 | - | - | BT_PHC1[5:3] | | | BT_PHC1[2:0] | | | 17H |
| | 0 | 1 | PHC2EN | - | BT_PHC2[5:3] | | | BT_PHC2[2:0] | | | 17H |

**BT_PHA[7:6]:** Soft start period of phase A.

**00b: 10mS** 01b: 20mS 10b: 30mS 11b: 40mS

**BT_PHA[5:3]:** Driving strength of phase A

000b: strength 1 001b: strength 2 **010b: strength 3** 011b: strength 4
100b: strength 5 101b: strength 6 110b: strength 7 111b: strength 8 (strongest)

**BT_PHA[2:0]:** Minimum OFF time setting of GDR in phase A

000b: 0.27uS 001b: 0.34uS 010b: 0.40uS 011b: 0.54uS
100b: 0.80uS 101b: 1.54uS 110b: 3.34uS **111b: 6.58uS**

**BT_PHB[7:6]:** Soft start period of phase B.

**00b: 10mS** 01b: 20mS 10b: 30mS 11b: 40mS

**BT_PHB[5:3]:** Driving strength of phase B

000b: strength 1 001b: strength 2 **010b: strength 3** 011b: strength 4
100b: strength 5 101b: strength 6 110b: strength 7 111b: strength 8 (strongest)

**BT_PHB[2:0]:** Minimum OFF time setting of GDR in phase B

000b: 0.27uS 001b: 0.34uS 010b: 0.40uS 011b: 0.54uS
100b: 0.80uS 101b: 1.54uS 110b: 3.34uS **111b: 6.58uS**

**BT_PHC1[5:3]:** Driving strength of phase C1

| | | | |
|---|---|---|---|
| 000b: strength 1 | 001b: strength 2 | **010b: strength 3** | 011b: strength 4 |
| 100b: strength 5 | 101b: strength 6 | 110b: strength 7 | 111b: strength 8 (strongest) |

**BT_PHC1[2:0]:** Minimum OFF time setting of GDR in phase C1

| | | | |
|---|---|---|---|
| 000b: 0.27uS | 001b: 0.34uS | 010b: 0.40uS | 011b: 0.54uS |
| 100b: 0.80uS | 101b: 1.54uS | 110b: 3.34uS | **111b: 6.58uS** |

**PHC2EN:** **Booster phase-C2 enable**

0: Booster phase-C2 disable
Phase-C1 setting always is applied for booster phase-C.

1: Booster phase-C2 enable
If temperature > temperature boundary phase-C2(RE7h[7:0]), phase-C1 setting is applied for booster phase-C.
If temperature <= temperature boundary phase-C2(RE7h[7:0]), phase-C2 setting is applied for booster phase-C.

**BT_PHC2[5:3]:** Driving strength of phase C2

| | | | |
|---|---|---|---|
| 000b: strength 1 | 001b: strength 2 | **010b: strength 3** | 011b: strength 4 |
| 100b: strength 5 | 101b: strength 6 | 110b: strength 7 | 111b: strength 8 (strongest) |

**BT_PHC2[2:0]:** Minimum OFF time setting of GDR in phase C2

| | | | |
|---|---|---|---|
| 000b: 0.27uS | 001b: 0.34uS | 010b: 0.40uS | 011b: 0.54uS |
| 100b: 0.80uS | 101b: 1.54uS | 110b: 3.34uS | **111b: 6.58uS** |

### (8) DEEP SLEEP (DSLP) (R07H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Deep Sleep | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 07H |
| | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | A5H |

After this command is transmitted, the chip will enter Deep Sleep Mode to save power. Deep Sleep Mode will return to Standby Mode by hardware reset. The only one parameter is a check code, the command will be executed if check code = 0xA5.

### (9) DATA START TRANSMISSION 1 (DTM1) (R10H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Starting data transmission | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 10H |
| | 0 | 1 | Pixel1 | Pixel2 | Pixel3 | Pixel4 | Pixel5 | Pixel6 | Pixel7 | Pixel8 | -- |
| | 0 | 1 | : | : | : | : | : | : | : | : | -- |
| | 0 | 1 | Pixel(n-7) | Pixel(n-6) | Pixel(n-5) | Pixel(n-4) | Pixel(n-3) | Pixel(n-2) | Pixel(n-1) | Pixel(n) | -- |

This command starts transmitting data and write them into SRAM.

In KW mode, this command writes "OLD" data to SRAM.

In KWR mode, this command writes "K/W" data to SRAM.

In Program mode, this command writes "OTP" data to SRAM for programming.

### (10) DATA STOP (DSP) (R11H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Stopping data transmission | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 11H |
| | 1 | 1 | data_flag | - | - | - | - | - | - | - | 00H |

Check the completeness of data. If data is complete, start to refresh display.

**Data_flag:** Data flag of receiving user data.

0: Driver didn't receive all the data.
1: Driver has already received all the one-frame data (DTM1 and DTM2).

After "Data Start" (R10h) or "Data Stop" (R11h) commands and when data_flag=1, the refreshing of panel starts and BUSY_N signal will become "0".

### (11) DISPLAY REFRESH (DRF) (R12H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Refreshing the display | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 12H |

While user sent this command, driver will refresh display (data/VCOM) according to SRAM data and LUT.

After Display Refresh command, BUSY_N signal will become "0" and the refreshing of panel starts.

### (12) DATA START TRANSMISSION 2 (DTM2) (R13H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Starting data transmission | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 13H |
| | 0 | 1 | Pixel1 | Pixel2 | Pixel3 | Pixel4 | Pixel5 | Pixel6 | Pixel7 | Pixel8 | -- |
| | 0 | 1 | : | : | : | : | : | : | : | : | -- |
| | 0 | 1 | Pixel(n-7) | Pixel(n-6) | Pixel(n-5) | Pixel(n-4) | Pixel(n-3) | Pixel(n-2) | Pixel(n-1) | Pixel(n) | -- |

This command starts transmitting data and write them into SRAM.

In KW mode, this command writes "NEW" data to SRAM.

In KWR mode, this command writes "RED" data to SRAM.

### (13) DUAL SPI MODE (DUSPI) (R15H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Stopping data transmission | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 15H |
| | 0 | 1 | - | - | MM_EN | DUSPI_EN | - | - | - | - | 00H |

This command sets dual SPI mode.

**MM_EN:** MM input pin definition enable.

0: MM input pin definition disable
1: MM input pin definition enable.

**DUSPI_EN:** Dual SPI mode enable.

0: Dual SPI mode disable (single SPI mode)
1: Dual SPI mode enable

### (14) AUTO SEQUENCE (AUTO) (R17H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Auto Sequence | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 17H |
| | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | A5H |

The command can enable the internal sequence to execute several commands continuously. The successive execution can minimize idle time to avoid unnecessary power consumption and reduce the complexity of host's control procedure. The sequence contains several operations, including PON, DRF, POF, DSLP.

AUTO (0x17) + Code(0xA5) = (PON → DRF → POF)

AUTO (0x17) + Code(0xA7) = (PON → DRF → POF → DSLP)

GOODISPLAY

### (15) VCOM LUT (LUTC) (R20H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Build Look-up Table for VCOM (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 20H |
| | 0 | 1 | LEVEL SELECT-0 | | LEVEL SELECT-1 | | LEVEL SELECT-2 | | LEVEL SELECT-3 | | -- |
| | 0 | 1 | NUMBER OF FRAMES-0 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-1 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-2 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-3 | | | | | | | | -- |
| | 0 | 1 | TIMES TO REPEAT | | | | | | | | -- |

This command stores VCOM Look-Up Table with 10 groups of data. Each group contains information for one state and is stored with 6 bytes (byte 2~7, 8~13, 14~19, 20~25, …), while the sixth byte indicates how many times that phase will repeat.

**Bytes 2, 8, 14, 20, 26, 32, 38, 44, 50, 56:**

D[7:6], D[5:4], D[3:2], D[1:0]: Level Selection

00b: VCOM_DC
01b: VDH+VCOM_DC (VCOMH)
10b: VDL+VCOM_DC (VCOML)
11b: Floating

**Bytes 3~6, 9~12, 15~18, 21~24, 27~30, 33~36, 39~42, 45~48, 51~54, 57~60:**

Number of Frames

0000 0000b: 0 frame
: :
1111 1111b: 255 frames

**Bytes 7, 13, 19, 25, 31, 37, 43, 49, 55, 61:**

Times to Repeat

0000 0000b: 0 time
: :
1111 1111b: 255 times

If KW/R=0 (KWR mode), all 10 groups are used.
If KW/R=1 (KW mode), only 7 groups are used.

### (16) W2W LUT (LUTWW) (R21H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Build White Look-up Table for W2W (43-byte command, structure of bytes 2~7 repeated 7 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 21H |
| | 0 | 1 | LEVEL SELECT-0 | | LEVEL SELECT-1 | | LEVEL SELECT-2 | | LEVEL SELECT-3 | | -- |
| | 0 | 1 | NUMBER OF FRAMES-0 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-1 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-2 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-3 | | | | | | | | -- |
| | 0 | 1 | TIMES TO REPEAT | | | | | | | | -- |

This command stores White-to-White Look-Up Table with 7 groups of data. Each group contains information for one state and is stored with 6 bytes (byte 2~7, 8~13, 14~19, 20~25, …), while the sixth byte indicates how many times that phase will repeat.

**Bytes 2, 8, 14, 20, 26, 32, 38:**

Level Selection.

00b: GND
01b: VDH
10b: VDL
11b: VDHR

**Bytes 3~6, 9~12, 15~18, 21~24, 27~30, 33~36, 39~42:**

Number of Frames

0000 0000b: 0 frame
: :
1111 1111b: 255 frames

**Bytes 7, 13, 19, 25, 31, 37, 43:**

Times to Repeat

0000 0000b: 0 time
: :
1111 1111b: 255 times

If KW/R=0 (KWR mode), LUTWW is not used.
If KW/R=1 (KW mode), LUTWW is used.

GOODISPLAY

### (17) K2W LUT (LUTKW / LUTR) (R22H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Build Look-up Table for K2W or Red (61-byte command, structure of bytes 2~7 repeated 10 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 22H |
| | 0 | 1 | LEVEL SELECT-0 | | LEVEL SELECT-1 | | LEVEL SELECT-2 | | LEVEL SELECT-3 | | -- |
| | 0 | 1 | NUMBER OF FRAMES-0 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-1 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-2 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-3 | | | | | | | | -- |
| | 0 | 1 | TIMES TO REPEAT | | | | | | | | -- |

This command stores White-to-White Look-Up Table with 10 groups of data. Each group contains information for one state and is stored with 6 bytes (byte 2~7, 8~13, 14~19, 20~25, …), while the sixth byte indicates how many times that phase will repeat.

**Bytes 2, 8, 14, 20, 26, 32, 38, 44, 50, 56:**

Level Selection.

00b: GND
01b: VDH
10b: VDL
11b: VDHR

**Bytes 3~6, 9~12, 15~18, 21~24, 27~30, 33~36, 39~42, 45~48, 51~54, 57~60:**

Number of Frames

0000 0000b: 0 frame
: :
: :
1111 1111b: 255 frames

**Bytes 7, 13, 19, 25, 31, 37, 43, 49, 55, 61:**

Times to Repeat

0000 0000b: 0 time
: :
: :
1111 1111b: 255 times

If KW/R=0 (KWR mode), all 10 groups are used.

If KW/R=1 (KW mode), only 7 groups are used.

### (18) W2K LUT (LUTWK / LUTW) (R23H)

This command builds Look-up Table for White-to-Black. Please refer to K2W LUT (LUTKW/LUTR) for similar definition details.

Regardless of KW/R=0 or KW/R=1, LUTWK/LUTW is used.

### (19) K2K LUT (LUTKK / LUTK) (R24H)

This command builds Look-up Table for Black-to-Black. Please refer to K2W LUT (LUTKW/LUTR) for similar definition details.

Regardless of KW/R=0 or KW/R=1, LUTKK/LUTK is used.

GOODISPLAY

### (20) BORDER LUT (LUTBD) (R25H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Build Look-up Table for Border (43-byte command, Bytes 2~7 repeated 7 times) | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 25H |
| | 0 | 1 | LEVEL SELECT-0 | | LEVEL SELECT-1 | | LEVEL SELECT-2 | | LEVEL SELECT-3 | | -- |
| | 0 | 1 | NUMBER OF FRAMES-0 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-1 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-2 | | | | | | | | -- |
| | 0 | 1 | NUMBER OF FRAMES-3 | | | | | | | | -- |
| | 0 | 1 | TIMES TO REPEAT | | | | | | | | -- |

This command stores White-to-White Look-Up Table with 7 groups of data. Each group contains information for one state and is stored with 6 bytes (byte 2~7, 8~13, 14~19, 20~25, …), while the sixth byte indicates how many times that phase will repeat.

**Bytes 2, 8, 14, 20, 26, 32, 38:**

Level selection.

| | | | | |
|---|---|---|---|---|
| BD_EN=0: | 00b: VCOM | 01b: VDH | 10b: VDL | 11b: VDHR |
| BD_EN=1: | 00b: VCOM | 01b: VBH(VCOM-VDL) | 10b: VBL(VCOM-VDH) | 11b: VDHR |

**Bytes 3~6, 9~12, 15~18, 21~24, 27~30, 33~36, 39~42:**

Number of Frames

0000 0000b: 0 frame

: :

: :

1111 1111b: 255 frames

**Bytes 7, 13, 19, 25, 31, 37, 43:**

Times to Repeat

0000 0000b: 0 time

: :

: :

1111 1111b: 255 times

Only 7 LUTBD groups are used in KW mode or KWR mode.

### (21) LUT OPTION (LUTOPT) (R2AH)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LUT Option | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 2AH |
| | 0 | 1 | STATE_XON[9:8] | | - | - | - | - | - | - | 00H |
| | 0 | 1 | STATE_XON[7:0] | | | | | | | | 00H |

This command sets XON control enable.

**STATE_XON[9:0]:**

All Gate ON (Each bit controls one state, STATE_XON [0] for state-1, STATE_XON [1] for state-2 …..)

00 0000 0000b: no All-Gate-ON

00 0000 0001b: State-1 All-Gate-ON

00 0000 0011b: State-1 and State2 All-Gate-ON

GOODISPLAY

### (22) KW LUT OPTION (KWOPT) (R2BH)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| KW LUT Option | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 2BH |
| | 0 | 1 | - | - | - | - | - | - | ATRED | NORED | 00H |
| | 0 | 1 | KWE[9:8] | | - | - | - | - | - | - | 00H |
| | 0 | 1 | KWE[7:0] | | | | | | | | 00H |

This command sets KW LUT mechanism option in KWR mode's LUT and only valid in K/W/R mode.

**{ATRED, NORED}:** KW LUT or KWR LUT selection control

| ATRED | NORED | Description |
|---|---|---|
| 0 | 0 | KWR LUT always |
| 0 | 1 | KW LUT only |
| 1 | 0 | Auto detect by red data |
| 1 | 1 | KW LUT only |

**KWE[9:0]:**

KW LUT enable control bits. Each bit controls one state, KWE[0] for state-1, KWE[1] for state-2, … .

At least 1 Enable Control bit should be set when KW LUT only is selected in KWR mode.

00 0000 0001b: KW LUT enable in State-1

00 0000 0011b: KW LUT enable in State-1 and State2

00 0000 1011b: KW LUT enable in State-1, State2 and State-4

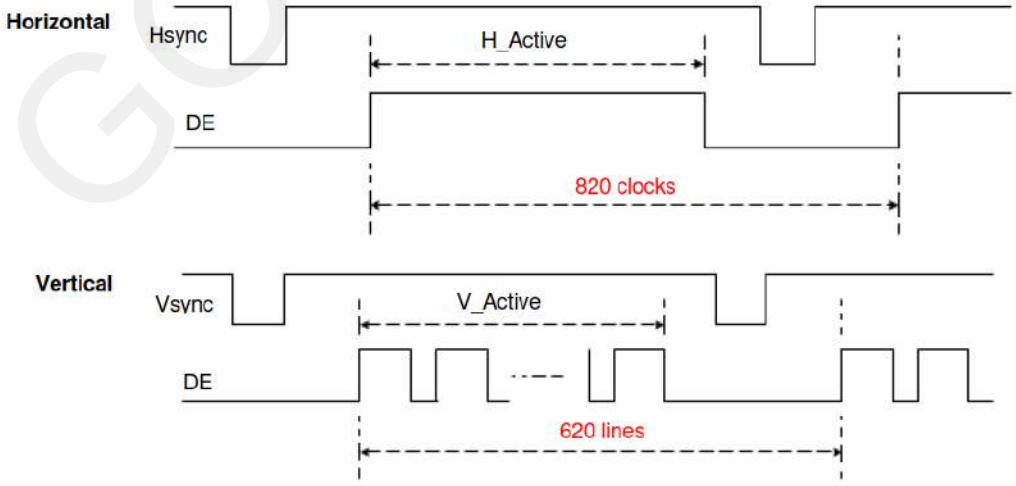
### (23) PLL CONTROL (PLL) (R30H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Controlling PLL | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 30H |
| | 0 | 1 | - | - | - | - | FRS[3:0] | | | | 06H |

The command controls the PLL clock frequency. The PLL structure must support the following frame rates:

**FMR[3:0]:** Frame rate setting

| FRS | Frame rate | FRS | Frame rate |
|---|---|---|---|
| 0000 | 5Hz | 1000 | 70Hz |
| 0001 | 10Hz | 1001 | 80Hz |
| 0010 | 15Hz | 1010 | 90Hz |
| 0011 | 20Hz | 1011 | 100Hz |
| 0100 | 30Hz | 1100 | 110Hz |
| 0101 | 40Hz | 1101 | 130Hz |
| **0110** | **50Hz** | 1110 | 150Hz |
| 0111 | 60Hz | 1111 | 200Hz |



### (24) TEMPERATURE SENSOR CALIBRATION (TSC) (R40H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sensing Temperature | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 40H |
| | 1 | 1 | D10/TS7 | D9/TS6 | D8/TS5 | D7/TS4 | D6 / TS3 | D5 / TS2 | D4 / TS1 | D3 / TS0 | 00H |
| | 1 | 1 | D2 | D1 | D0 | - | - | - | - | - | 00H |

This command enables internal or external temperature sensor, and reads the result.

**TS[7:0]:** When TSE (R41h) is set to 0, this command reads internal temperature sensor value.

**D[10:0]:** When TSE (R41h) is set to 1, this command reads external LM75 temperature sensor value.

| TS[7:0]/D[10:3] | Temp. (°C) |
|---|---|
| 1110_0111 | -25 |
| 1110_1000 | -24 |
| 1110_1001 | -23 |
| 1110_1010 | -22 |
| 1110_1011 | -21 |
| 1110_1100 | -20 |
| 1110_1101 | -19 |
| 1110_1110 | -18 |
| 1110_1111 | -17 |
| 1111_0000 | -16 |
| 1111_0001 | -15 |
| 1111_0010 | -14 |
| 1111_0011 | -13 |
| 1111_0100 | -12 |
| 1111_0101 | -11 |
| 1111_0110 | -10 |
| 1111_0111 | -9 |
| 1111_1000 | -8 |
| 1111_1001 | -7 |
| 1111_1010 | -6 |
| 1111_1011 | -5 |
| 1111_1100 | -4 |
| 1111_1101 | -3 |
| 1111_1110 | -2 |
| 1111_1111 | -1 |

| TS[7:0]/D[10:3] | Temp. (°C) |
|---|---|
| 0000_0000 | 0 |
| 0000_0001 | 1 |
| 0000_0010 | 2 |
| 0000_0011 | 3 |
| 0000_0100 | 4 |
| 0000_0101 | 5 |
| 0000_0110 | 6 |
| 0000_0111 | 7 |
| 0000_1000 | 8 |
| 0000_1001 | 9 |
| 0000_1010 | 10 |
| 0000_1011 | 11 |
| 0000_1100 | 12 |
| 0000_1101 | 13 |
| 0000_1110 | 14 |
| 0000_1111 | 15 |
| 0001_0000 | 16 |
| 0001_0001 | 17 |
| 0001_0010 | 18 |
| 0001_0011 | 19 |
| 0001_0100 | 20 |
| 0001_0101 | 21 |
| 0001_0110 | 22 |
| 0001_0111 | 23 |
| 0001_1000 | 24 |

| TS[7:0]/D[10:3] | Temp. (°C) |
|---|---|
| 0001_1001 | 25 |
| 0001_1010 | 26 |
| 0001_1011 | 27 |
| 0001_1100 | 28 |
| 0001_1101 | 29 |
| 0001_1110 | 30 |
| 0001_1111 | 31 |
| 0010_0000 | 32 |
| 0010_0001 | 33 |
| 0010_0010 | 34 |
| 0010_0011 | 35 |
| 0010_0100 | 36 |
| 0010_0101 | 37 |
| 0010_0110 | 38 |
| 0010_0111 | 39 |
| 0010_1000 | 40 |
| 0010_1001 | 41 |
| 0010_1010 | 42 |
| 0010_1011 | 43 |
| 0010_1100 | 44 |
| 0010_1101 | 45 |
| 0010_1110 | 46 |
| 0010_1111 | 47 |
| 0011_0000 | 48 |
| 0011_0001 | 49 |

### (25) TEMPERATURE SENSOR ENABLE (TSE) (R41H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Enable Temperature Sensor /Offset | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 41H |
| | 0 | 1 | TSE | - | - | - | TO[3:0] | | | | 00H |

This command selects Internal or External temperature sensor.

**TSE:** Internal temperature sensor switch

**0: Enable (default)** 1: Disable; using external sensor.

**TO[3:0]:** Temperature offset.

| TO[3:0] | Calibration |
|---|---|
| **0000 b** | **+0 (Default)** |
| 0001 | +1 |
| 0010 | +2 |
| 0011 | +3 |
| 0100 | +4 |
| 0101 | +5 |
| 0110 | +6 |
| 0111 | +7 |

| TO[3:0] | Calibration |
|---|---|
| 1000 | -8 |
| 1001 | -7 |
| 1010 | -6 |
| 1011 | -5 |
| 1100 | -4 |
| 1101 | -3 |
| 1110 | -2 |
| 1111 | -1 |

### (26) TEMPERATURE SENSOR WRITE (TSW) (R42H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Write External Temperature Sensor | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 42H |
| | 0 | 1 | WATTR[7:0] | | | | | | | | 00H |
| | 0 | 1 | WMSB[7:0] | | | | | | | | 00H |
| | 0 | 1 | WLSB[7:0] | | | | | | | | 00H |

This command writes the temperature sensed by the temperature sensor.

**WATTR[7:6]:** I²C Write Byte Number

00b : 1 byte (head byte only)
01b : 2 bytes (head byte + pointer)
10b : 3 bytes (head byte + pointer + 1st parameter)
11b : 4 bytes (head byte + pointer + 1st parameter + 2nd parameter)

**WATTR[5:3]:** User-defined address bits (A2, A1, A0)

**WATTR[2:0]:** Pointer setting

**WMSB[7:0]:** MSByte of write-data to external temperature sensor

**WLSB[7:0]:** LSByte of write-data to external temperature sensor

### (27) TEMPERATURE SENSOR READ (TSR) (R43H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Read External Temperature Sensor | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 43H |
| | 1 | 1 | RMSB[7:0] | | | | | | | | 00H |
| | 1 | 1 | RLSB[7:0] | | | | | | | | 00H |

This command reads the temperature sensed by the temperature sensor.

**RMSB[7:0]:** MSByte read data from external temperature sensor

**RLSB[7:0]:** LSByte read data from external temperature sensor

### (28) PANEL GLASS CHECK (PBC)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Check Panel Glass | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 44H |
| | 1 | 1 | - | - | - | - | - | - | - | PSTA | 00H |

This command is used to enable panel check, and to disable after reading result.

**PSTA:** 0: Panel check fail (panel broken) 1: Panel check pass

### (29) VCOM AND DATA INTERVAL SETTING (CDI) (R50H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Interval between VCOM and Data | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 50h |
| | 0 | 1 | BDZ | - | BDV[1:0] | | N2OCP | - | DDX[1:0] | | 31h |
| | 0 | 1 | - | - | - | SDEND | CDI[3:0] | | | | 07H |

This command indicates the interval of VCOM and data output. When setting the vertical back porch, the total blanking will be kept (20 Hsync).

**BDZ:** Border Hi-Z control

**0: Border output Hi-Z disabled (default)** 1: Border output Hi-Z enabled

GOODISPLAY

**BDV[1:0]:** Border LUT selection

KWR mode (KW/R=0)

| DDX[0] | BDV[1:0] | LUT |
|---|---|---|
| 0 | 00 | LUTBD |
| | 01 | LUTR |
| | 10 | LUTW |
| | 11 | LUTK |
| **1 (Default)** | 00 | LUTK |
| | 01 | LUTW |
| | 10 | LUTR |
| | 11 | LUTBD |

KW mode (KW/R=1)

| DDX[0] | BDV[1:0] | LUT |
|---|---|---|
| 0 | 00 | LUTBD |
| | 01 | LUTKW (1 → 0) |
| | 10 | LUTWK (0 → 1) |
| | 11 | LUTKK (0 → 0) |
| **1 (Default)** | 00 | LUTKK (0 → 0) |
| | 01 | LUTWK (1 → 0) |
| | 10 | LUTKW (0 → 1) |
| | 11 | LUTBD |

**N2OCP:** Copy frame data from NEW data to OLD data enable control after display refresh with NEW/OLD in KW mode.

**0: Copy NEW data to OLD data disabled (default)** 1: Copy NEW data to OLD data enabled

**DDX[1:0]:** Data polarity.

Under KWR mode (KW/R=0):

DDX[1] is for RED data.
DDX[0] is for K/W data,

| DDX[1:0] | Data {Red, K/W} | LUT |
|---|---|---|
| 00 | 00 | LUTW |
| | 01 | LUTK |
| | 10 | LUTR |
| | 11 | LUTR |
| **01 (Default)** | 00 | LUTK |
| | 01 | LUTW |
| | 10 | LUTR |
| | 11 | LUTR |

| DDX[1:0] | Data {Red, K/W} | LUT |
|---|---|---|
| 10 | 00 | LUTR |
| | 01 | LUTR |
| | 10 | LUTW |
| | 11 | LUTK |
| 11 | 00 | LUTR |
| | 01 | LUTR |
| | 10 | LUTK |
| | 11 | LUTW |

Under KW mode (KW/R=1):

DDX[1]=0 is for KW mode with NEW/OLD,
DDX[1]=1 is for KW mode without NEW/OLD.

| DDX[1:0] | Data {NEW, OLD} | LUT |
|---|---|---|
| 00 | 00 | LUTWW (0 → 0) |
| | 01 | LUTKW (1 → 0) |
| | 10 | LUTWK (0 → 1) |
| | 11 | LUTKK (1 → 1) |
| **01 (Default)** | 00 | LUTKK (0 → 0) |
| | 01 | LUTWK (1 → 0) |
| | 10 | LUTKW (0 → 1) |
| | 11 | LUTWW (1 → 1) |

| DDX[1:0] | Data {NEW} | LUT |
|---|---|---|
| 10 | 0 | LUTKW (1 → 0) |
| | 1 | LUTWK (0 → 1) |
| 11 | 0 | LUTWK (1 → 0) |
| | 1 | LUTKW (0 → 1) |

**SDEND:** source driving ending

0: source driver channels output 2-frame 0V at the end

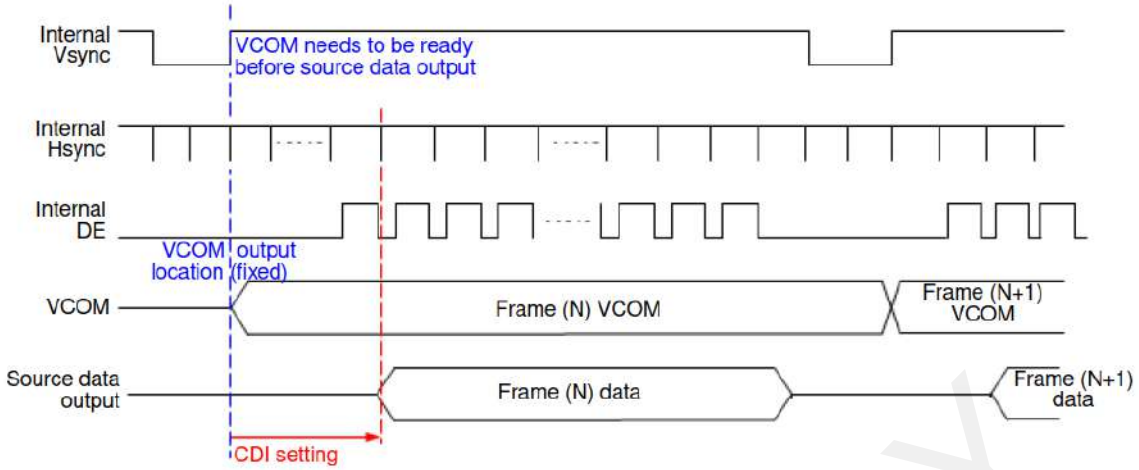
1: source driver channels keep the last state at the end

**CDI[3:0]:** VCOM and data interval

| CDI[3:0] | VCOM and Data Interval |
|---|---|
| 0000 b | 17 hsync |
| 0001 | 16 |
| 0010 | 15 |
| 0011 | 14 |
| 0100 | 13 |
| 0101 | 12 |
| 0110 | 11 |
| **0111** | **10 (Default)** |

| CDI[3:0] | VCOM and Data Interval |
|---|---|
| 1000 | 9 |
| 1001 | 8 |
| 1010 | 7 |
| 1011 | 6 |
| 1100 | 5 |
| 1101 | 4 |
| 1110 | 3 |
| 1111 | 2 |

GOODISPLAY



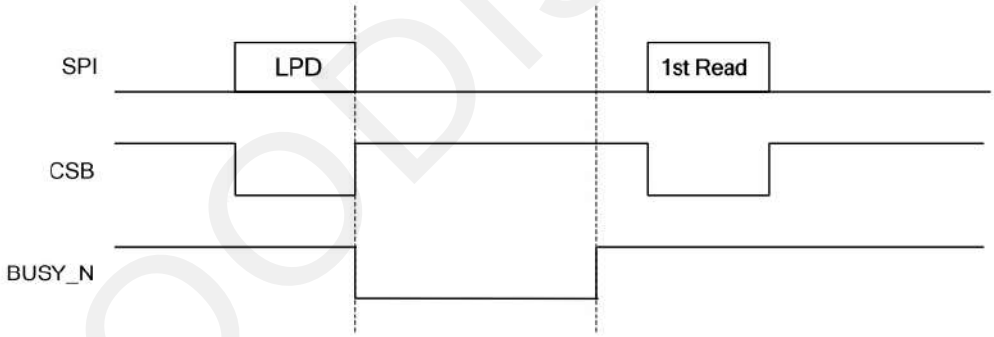

### (30) LOW POWER DETECTION (LPD) (R51H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Detect Low Power | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 51h |
| | 1 | 1 | - | - | - | - | - | - | - | LPD | 01h |

This command indicates the input power condition. Host can read this flag to learn the battery condition.

**LPD:** Internal Low Power Detection Flag

0: Low power input ($V_{DD}$ < 2.5V, 2.4V, 2.3V, or 2.2V, selected by LVD_SEL[1:0] in command LVSEL)
**1: Normal status (default)**




### (31) END VOLTAGE SETTING (EVS) (R52H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| End Voltage Setting | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 52h |
| | 0 | 1 | - | - | - | - | VCEND | - | BDEND[1:0] | | 02h |

This command selects source end voltage and border end voltage after LUTs are finished.

**VCEND:** VCOM end voltage selection

**0b: VCOM_DC** 1b: floating

**BDEND[1:0]:** Border end voltage selection

00b: 0V 01b: 0V **10b: VCOM_DC** 11b: floating

GOODISPLAY

### (32) TCON SETTING (TCON) (R60H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Gate/Source Non-overlap Period | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 60h |
| | 0 | 1 | S2G[3:0] | | | | G2S[3:0] | | | | 22h |

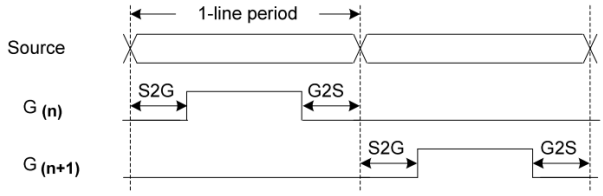
This command defines non-overlap period of Gate and Source.

**S2G[3:0] or G2S[3:0]:** Source to Gate / Gate to Source Non-overlap period

| S2G[3:0] or G2S[3:0] | Period |
|---|---|
| 0000 b | 4 |
| 0001 | 8 |
| **0010** | **12 (Default)** |
| 0011 | 16 |
| 0100 | 20 |
| 0101 | 24 |
| 0110 | 28 |
| 0111 | 32 |

| S2G[3:0] or G2S[3:0] | Period |
|---|---|
| 1000 b | 36 |
| 1001 | 40 |
| 1010 | 44 |
| 1011 | 48 |
| 1100 | 52 |
| 1101 | 56 |
| 1110 | 60 |
| 1111 | 64 |

Period Unit = 667 nS.




### (33) RESOLUTION SETTING (TRES) (R61H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Display Resolution | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 61h |
| | 0 | 1 | - | - | - | - | - | - | HRES[9:8] | | 03h |
| | 0 | 1 | HRES[7:3] | | | | | 0 | 0 | 0 | 20h |
| | 0 | 1 | - | - | - | - | - | - | VRES[9:8] | | 02h |
| | 0 | 1 | VRES[7:0] | | | | | | | | 58h |

This command defines resolution setting.

**HRES[9:3]:** Horizontal Display Resolution (Value range: 01h ~ 64h)

**VRES[9:0]:** Vertical Display Resolution (Value range: 001h ~ 258h)

Active channel calculation, assuming HST[9:0]=0, VST[9:0]=0:

Gate: First active gate = G0;
Last active gate = VRES[9:0] – 1

Source: First active source = S0;
Last active source = HRES[9:3]*8 – 1

Example: 128 (source) x 272 (gate), assuming HST[9:0]=0, VST[9:0]=0

Gate: First active gate = G0,
Last active gate = G271; (VRES[9:0] = 272, 272 – 1= 271)

Source: First active source = S0,
Last active source = S127; (HRES[9:3]=16, 16*8 – 1 = 127)

### (34) GATE/SOURCE START SETTING (GSST) (R65H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Gate/Source Start | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 65h |
| | 0 | 1 | - | - | - | - | - | - | HST[9:8] | | 00h |
| | 0 | 1 | HST[7:3] | | | | | 0 | 0 | 0 | 00h |
| | 0 | 1 | - | - | - | - | - | - | VST[9:8] | | 00h |
| | 0 | 1 | VST[7:0] | | | | | | | | 00h |

This command defines resolution start gate/source position.

**HST[9:3]:** Horizontal Display Start Position (Source). (Value range: 00h ~ 63h)

**VST[9:0]:** Vertical Display Start Position (Gate). (Value range: 000h ~ 257h)

Example : For 128(Source) x 240(Gate)
HST[9:3] = 4 (HST[9:0] = 4*8 = 32),
VST[9:0] = 32

Gate: First active gate = G32 (VST[9:0] = 32),
Last active gate = G271 (VRES[9:0] = 240, VST[9:0] = 32, 240-1+32=271)

Source: First active source = S32 (HST[9:0]= 32),
Last active source = S239 (HRES[9:0] = 128, HST[9:0] = 32, 128-1+32=239)

GOODISPLAY

### (35) REVISION (REV) (R70H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LUT/Chip Revision | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 70h |
| | 1 | 1 | PROD_REV[23:16] | | | | | | | | FFh |
| | 1 | 1 | PROD_REV[15:8] | | | | | | | | FFh |
| | 1 | 1 | PROD_REV[7:0] | | | | | | | | FFh |
| | 1 | 1 | LUT_REV[23:16] | | | | | | | | FFh |
| | 1 | 1 | LUT_REV[15:8] | | | | | | | | FFh |
| | 1 | 1 | LUT_REV[7:0] | | | | | | | | FFh |
| | 1 | 1 | CHIP_REV[7:0] | | | | | | | | 0Ch |

The command reads the product revision, LUT revision and chip revision.

**PROD_REV[23:0]:** Product Revision. PROD_REV[23:0] is read from OTP address 0x0BDD ~ 0X0BDF or 0x17DD ~ 0x17DF.

**LUT_REV[23:0]:** LUT Revision. LUT_REV[23:0] is read from OTP address 0x0BE0 ~ 0X0BE2 or 0x17E0.~ 0x17E2.

**CHIP_REV[7:0]:** Chip Revision, fixed at 00001100b.

### (36) GET STATUS (FLG) (R71H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Read Flags | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 71h |
| | 1 | 1 | - | PTL_ Flag | I²C_ERR | I²C_ BUSYN | Data_ Flag | PON | POF | BUSY_N | 13h |

This command reads the IC status.

**PTL_Flag:** Partial display status (high: partial mode)

**I²C_ERR:** I²C master error status

**I²C_BUSYN:** I²C master busy status (low active)

**Data_Flag:** Driver has already received all the one frame data

**PON:** Power ON status

**POF:** Power OFF status

**BUSY_N:** Driver busy status (low active)

### (37) AUTO MEASURE VCOM (AMV) (R80H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Automatically measure VCOM | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 80h |
| | 0 | 1 | - | - | AMVT[1:0] | | XON | AMVS | AMV | AMVE | 10h |

This command triggers auto VCOM sensing mechanism.

**AMVT[1:0]:** Auto Measure VCOM Time

00b: 3s
10b: 8s
**01b: 5s (default)**
11b: 10s

**XON:** All Gate ON of AMV

**0: Gate normally scan during Auto Measure VCOM period. (default)**
1: All Gate ON during Auto Measure VCOM period.

**AMVS:** Source output of AMV

**0: Source output 0V during Auto Measure VCOM period. (default)**
1: Source output VDHR during Auto Measure VCOM period.

**AMV:** Analog signal

**0: Get VCOM value with the VV command (R81h) (default)**
1: Get VCOM value in analog signal. (External analog to digital converter)

**AMVE:** Auto Measure VCOM Enable (/Disable)

**0: No effect (default)**
1: Trigger auto VCOM sensing.

GOODISPLAY

### (38) VCOM VALUE (VV) (R81H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Automatically measure VCOM | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 81h |
| | 1 | 1 | - | VV[6:0] | | | | | | | 00h |

This command gets the VCOM value.

**VV[6:0]:** VCOM Value Output

| VV [6:0] | VCOM Voltage (V) | VV [6:0] | VCOM Voltage (V) | VV [6:0] | VCOM Voltage (V) |
|---|---|---|---|---|---|
| 000 0000b | -0.10 | 001 1011b | -1.45 | 011 0110b | -2.80 |
| 000 0001b | -0.15 | 001 1100b | -1.50 | 011 0111b | -2.85 |
| 000 0010b | -0.20 | 001 1101b | -1.55 | 011 1000b | -2.90 |
| 000 0011b | -0.25 | 001 1110b | -1.60 | 011 1001b | -2.95 |
| 000 0100b | -0.30 | 001 1111b | -1.65 | 011 1010b | -3.00 |
| 000 0101b | -0.35 | 010 0000b | -1.70 | 011 1011b | -3.05 |
| 000 0110b | -0.40 | 010 0001b | -1.75 | 011 1100b | -3.10 |
| 000 0111b | -0.45 | 010 0010b | -1.80 | 011 1101b | -3.15 |
| 000 1000b | -0.50 | 010 0011b | -1.85 | 011 1110b | -3.20 |
| 000 1001b | -0.55 | 010 0100b | -1.90 | 011 1111b | -3.25 |
| 000 1010b | -0.60 | 010 0101b | -1.95 | 100 0000b | -3.30 |
| 000 1011b | -0.65 | 010 0110b | -2.00 | 100 0001b | -3.35 |
| 000 1100b | -0.70 | 010 0111b | -2.05 | 100 0010b | -3.40 |
| 000 1101b | -0.75 | 010 1000b | -2.10 | 100 0011b | -3.45 |
| 000 1110b | -0.80 | 010 1001b | -2.15 | 100 0100b | -3.50 |
| 000 1111b | -0.85 | 010 1010b | -2.20 | 100 0101b | -3.55 |
| 001 0000b | -0.90 | 010 1011b | -2.25 | 100 0110b | -3.60 |
| 001 0001b | -0.95 | 010 1100b | -2.30 | 100 0111b | -3.65 |
| 001 0010b | -1.00 | 010 1101b | -2.35 | 100 1000b | -3.70 |
| 001 0011b | -1.05 | 010 1110b | -2.40 | 100 1001b | -3.75 |
| 001 0100b | -1.10 | 010 1111b | -2.45 | 100 1010b | -3.80 |
| 001 0101b | -1.15 | 011 0000b | -2.50 | 100 1011b | -3.85 |
| 001 0110b | -1.20 | 011 0001b | -2.55 | 100 1100b | -3.90 |
| 001 0111b | -1.25 | 011 0010b | -2.60 | 100 1101b | -3.95 |
| 001 1000b | -1.30 | 011 0011b | -2.65 | 100 1110b | -4.00 |
| 001 1001b | -1.35 | 011 0100b | -2.70 | 100 1111b | -4.05 |
| 001 1010b | -1.40 | 011 0101b | -2.75 | | |

### (39) VCOM_DC SETTING (VDCS) (R82H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set VCOM_DC | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 82h |
| | 0 | 1 | - | VDCS[6:0] | | | | | | | 00h |

This command sets VCOM_DC value

**VDCS[6:0]:** VCOM_DC Setting

| VDCS [6:0] | VCOM Voltage (V) | VDCS [6:0] | VCOM Voltage (V) | VDCS [6:0] | VCOM Voltage (V) |
|---|---|---|---|---|---|
| 000 0000b | -0.10 | 001 1011b | -1.45 | 011 0110b | -2.80 |
| 000 0001b | -0.15 | 001 1100b | -1.50 | 011 0111b | -2.85 |
| 000 0010b | -0.20 | 001 1101b | -1.55 | 011 1000b | -2.90 |
| 000 0011b | -0.25 | 001 1110b | -1.60 | 011 1001b | -2.95 |
| 000 0100b | -0.30 | 001 1111b | -1.65 | 011 1010b | -3.00 |
| 000 0101b | -0.35 | 010 0000b | -1.70 | 011 1011b | -3.05 |
| 000 0110b | -0.40 | 010 0001b | -1.75 | 011 1100b | -3.10 |
| 000 0111b | -0.45 | 010 0010b | -1.80 | 011 1101b | -3.15 |
| 000 1000b | -0.50 | 010 0011b | -1.85 | 011 1110b | -3.20 |
| 000 1001b | -0.55 | 010 0100b | -1.90 | 011 1111b | -3.25 |
| 000 1010b | -0.60 | 010 0101b | -1.95 | 100 0000b | -3.30 |
| 000 1011b | -0.65 | 010 0110b | -2.00 | 100 0001b | -3.35 |
| 000 1100b | -0.70 | 010 0111b | -2.05 | 100 0010b | -3.40 |
| 000 1101b | -0.75 | 010 1000b | -2.10 | 100 0011b | -3.45 |
| 000 1110b | -0.80 | 010 1001b | -2.15 | 100 0100b | -3.50 |
| 000 1111b | -0.85 | 010 1010b | -2.20 | 100 0101b | -3.55 |
| 001 0000b | -0.90 | 010 1011b | -2.25 | 100 0110b | -3.60 |
| 001 0001b | -0.95 | 010 1100b | -2.30 | 100 0111b | -3.65 |
| 001 0010b | -1.00 | 010 1101b | -2.35 | 100 1000b | -3.70 |
| 001 0011b | -1.05 | 010 1110b | -2.40 | 100 1001b | -3.75 |
| 001 0100b | -1.10 | 010 1111b | -2.45 | 100 1010b | -3.80 |
| 001 0101b | -1.15 | 011 0000b | -2.50 | 100 1011b | -3.85 |
| 001 0110b | -1.20 | 011 0001b | -2.55 | 100 1100b | -3.90 |
| 001 0111b | -1.25 | 011 0010b | -2.60 | 100 1101b | -3.95 |
| 001 1000b | -1.30 | 011 0011b | -2.65 | 100 1110b | -4.00 |
| 001 1001b | -1.35 | 011 0100b | -2.70 | 100 1111b | -4.05 |
| 001 1010b | -1.40 | 011 0101b | -2.75 | | |

### (40) PARTIAL WINDOW (PTL) (R90H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Partial Window | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 90h |
| | 0 | 1 | - | - | - | - | - | - | HRST[9:8] | | 00h |
| | 0 | 1 | HRST[7:3] | | | | | 0 | 0 | 0 | 00h |
| | 0 | 1 | - | - | - | - | - | - | HRED[9:8] | | 03h |
| | 0 | 1 | HRED[7:3] | | | | | 1 | 1 | 1 | 1Fh |
| | 0 | 1 | - | - | - | - | - | - | VRST[9:8] | | 00h |
| | 0 | 1 | VRST[7:0] | | | | | | | | 00h |
| | 0 | 1 | - | - | - | - | - | - | VRED[9:8] | | 02h |
| | 0 | 1 | VRED[7:0] | | | | | | | | 57h |
| | 0 | 1 | - | - | - | - | - | - | - | PT_SCAN | 01h |

This command sets partial window.

**HRST[9:3]:** Horizontal start channel bank. (Value range: 00h~63h)

**HRED[9:3]:** Horizontal end channel bank. (Value range: 00h~63h). HRED must be greater than HRST.

**VRST[9:0]:** Vertical start line. (Value range: 000h~257h)

**VRED[9:0]:** Vertical end line. (Value range: 000h~257h). VRED must be greater than VRST.

**PT_SCAN:** 0: Gates scan only inside of the partial window.
**1: Gates scan both inside and outside of the partial window. (default)**

### (41) PARTIAL IN (PTIN) (R91H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Partial In | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 91h |

This command makes the display enter partial mode.

### (42) PARTIAL OUT (PTOUT) (R92H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Partial Out | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 92h |

This command makes the display exit partial mode and enter normal mode.

### (43) PROGRAM MODE (PGM) (RA0H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Enter Program Mode | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | A0h |

After this command is issued, the chip would enter the program mode.

After the programming procedure completed, a hardware reset is necessary for leaving program mode.

### (44) ACTIVE PROGRAM (APG) (RA1H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Active Program OTP | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | A1h |

After this command is transmitted, the programming state machine would be activated.

The BUSY_N flag would fall to 0 until the programming is completed.

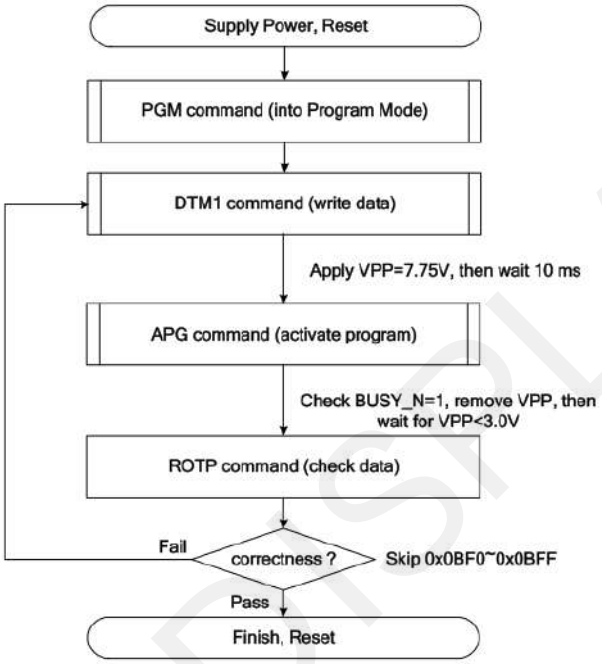
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### (45) READ OTP DATA (ROTP) (RA2H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Read OTP data for check | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | A2h |
| | 1 | 1 | The data of address 0x000 in the OTP | | | | | | | | -- |
| | 1 | 1 | The data of address 0x001 in the OTP | | | | | | | | -- |
| | 1 | 1 | : | | | | | | | | -- |
| | 1 | 1 | The data of address (n-1) in the OTP | | | | | | | | -- |
| | 1 | 1 | The data of address (n) in the OTP | | | | | | | | -- |

The command is used for reading the content of OTP for checking the data of programming.

The value of (n) is depending on the amount of programmed data, the max address = 0x17FF.




The sequence of programming OTP.

### (46) CASCADE SETTING (CCSET) (RE0H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Cascade Option | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | E0h |
| | 0 | 1 | - | - | - | - | - | - | TSFIX | CCEN | 00h |

This command is used for cascade.

**TSFIX:** Let the value of slave's temperature is same as the master's.

**0: Temperature value is defined by internal temperature sensor / external LM75. (default)**

1: Temperature value is defined by TS_SET[7:0] registers.

**CCEN:** Output clock enable/disable.

**0: Output 0V at CL pin. (default)**

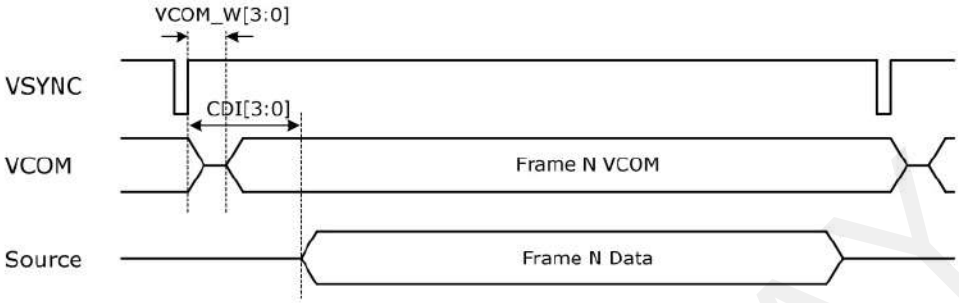
1: Output clock at CL pin to slave chip.

### (47) POWER SAVING (PWS) (RE3H)

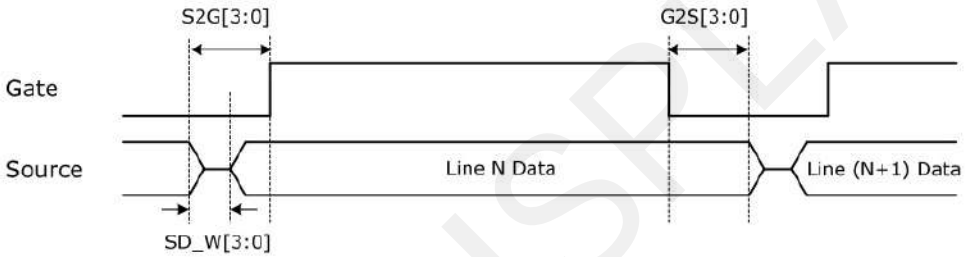
| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Power Saving for VCOM & Source | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | E3h |
| | 0 | 1 | VCOM_W[3:0] | | | | SD_W[3:0] | | | | 00h |

This command is set for saving power during refreshing period. If the output voltage of VCOM / Source is from negative to positive or from positive to negative, the power saving mechanism will be activated. The active period width is defined by the following two parameters.

**VCOM_W[3:0]:** VCOM power saving width (Unit: line period)




**SD_W[3:0]:** Source power saving width (Unit: 660nS)




### (48) LVD VOLTAGE SELECT (LVSEL) (RE4H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Select LVD Voltage | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | E4h |
| | 0 | 1 | - | - | - | - | - | - | LVD_SEL[1:0] | | 03h |

**LVD_SEL[1:0]:** Low Power Voltage selection

| LVD_SEL[1:0] | LVD value |
|---|---|
| 00 | < 2.2 V |
| 01 | < 2.3 V |
| 10 | < 2.4 V |
| 11 | < 2.5 V (default) |

### (49) FORCE TEMPERATURE (TSSET) (RE5H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Force Temperature Value for Cascade | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | E5h |
| | 0 | 1 | TS_SET[7:0] | | | | | | | | 00h |

This command is used for cascade to fix the temperature value of master and slave chip.

### (50) TEMPERATURE BOUNDARY PHASE-C2 (TSBDRY) (RE7H)

| Action | W/R | C/D | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Temperature Boundary Phase-C2 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | E7h |
| | 0 | 1 | TSBDRY_PHC2[7:0] | | | | | | | | 00h |

This command is used to set the temperature boundary to judge whether booster phase-C2 is applied or not.

## 8.Optical Specifications

Measurements are made with that the illumination is under an angle of 45 degree, the detection is perpendicular unless otherwise specified

| Symbol | Parameter | Conditions | Min | Typ. | Max | Units | Notes |
|---|---|---|---|---|---|---|---|
| R | White Reflectivity | White | 30 | 35 | - | % | 8-1 |
| CR | Contrast Ratio | Indoor | 8:1 | | - | | 8-2 |
| GN | 2Grey Level | - | | DS+(WS-DS)*n(m-1) | | | 8-3 |
| T update | Image update time | at 25 °C | | 3 | - | sec | |
| Life | | Topr | | 1000000times or 5years | | | |

**Notes:**
8-1. Luminance meter: Eye-One Pro Spectrophotometer.
8-2. CR=Surface Reflectance with all white pixel/Surface Reflectance with all black pixels.
8-3 WS: White state, DS: Dark state

## 9.Handling, Safety and Environment Requirements

| **Warning** |
|---|
| The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap. |
| **Caution** |
| The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. Disassembling the display module.<br>Disassembling the display module can cause permanent damage and invalidates the warranty agreements.<br>Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions. |
| **Data sheet status** |
| Product specification — This data sheet contains final product specifications. |
| **Limiting values** |
| Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134).Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability. |
| **Application information** |
| Where application information is given, it is advisory and does not form part of the specification. |

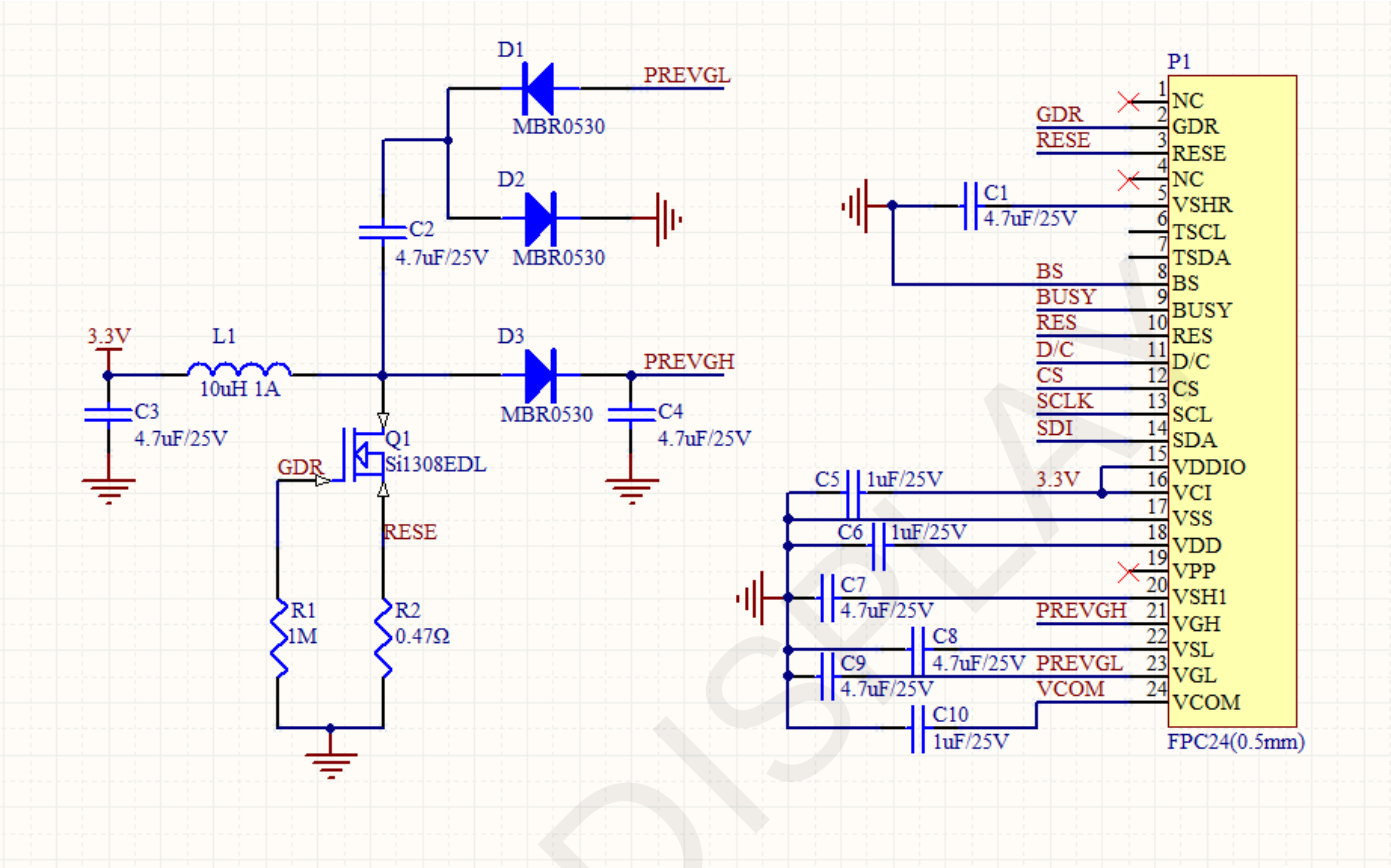
## 10.Reliability test

| NO | Test items | Test condition |
|---|---|---|
| 1 | Low-Temperature Storage | T = -25°C, 240 h<br>Test in white pattern |
| 2 | High-Temperature Storage | T=70° C，RH=40%，240h<br>Test in white pattern |
| 3 | High-Temperature Operation | T=50° C，RH=35%，240h |
| 4 | Low-Temperature Operation | 0° C，240h |
| 5 | High-Temperature, High-Humidity Operation | T=40° C，RH=80%，240h |
| 6 | High Temperature, High Humidity Storage | T=50° C，RH=80%, 240h<br>Test in white pattern |
| 7 | Temperature Cycle | 1 cycle:[-25° C 30min]→[+70 ° C 30 min] : 50 cycles<br>Test in white pattern |
| 8 | UV exposure Resistance | 765W/m² for 168hrs,40 °C<br>Test in white pattern |
| 9 | ESD Gun | Air+/-15KV;Contact+/-8KV<br>(Test finished product shell, not display only)<br>Air+/-8KV;Contact+/-6KV<br>(Naked EPD display, no including IC and FPC area)<br>Air+/-4KV;Contact+/-2KV<br>(Naked EPD display, including IC and FPC area) |

**Note:**
Put in normal temperature for 1hour after test finished, display performance is ok.

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## 11. Reference Circuit




| Part Name | Requirements for spare part |
|---|---|
| C1—C12 | 0603/0805; X5R/X7R;Voltage Rating:≥25V |
| R1、R2 | 0603/0805;1% variation,≥0.05W |
| D1—D3 | MBR0530: 1)Reverse DC Voltage≥30V 2)Io≥500mA 3)Forward voltage ≤430mV |
| Q1 | Si1308EDL:1)Drain-Source breakdown voltage≥30V 2)Vgs(th)≤1.5V 3)Rds(on)≤400mΩ |
| L1 | refer to NR3015: Io=500mA(max) |
| P1 | 24pins,0.5mm pitch |

## 12. Matched Development Kit

Our Development Kit designed for SPI E-paper Display aims to help users to learn how to use E-paper Display more easily. It can refresh black-white E-paper Display and three-color (black, white and red/Yellow) Good Display 's E-paper Display. And it is also added the functions of USB serial port, Raspberry Pi and LED indicator light ect.
DESPI Development Kit consists of the development board and the pinboard.

More details about the Development Kit, please click to the following link:

https://www.good-display.com/product/53/

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## 13. Inspection condition

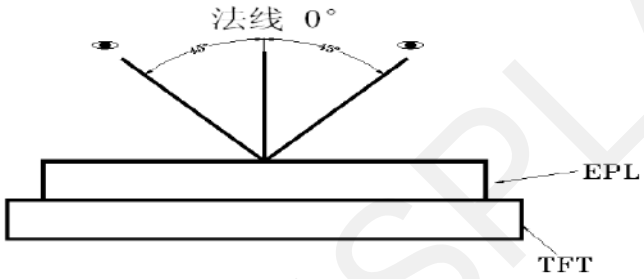
### 13. 1 Environment

Temperature： 25±3℃
Humidity： 55±10%RH

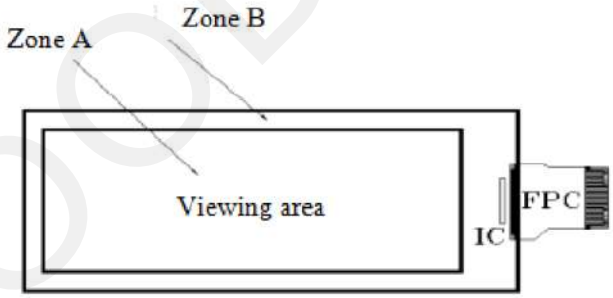
### 13. 2 Illuminance

Brightness:1200～1500LUX;distance:20-30CM;Angle:Relate 45°surround.

### 13.3 Inspection method



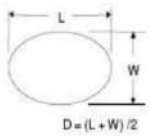

### 13. 4 Display area

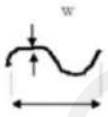


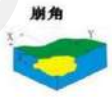
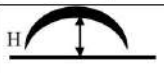

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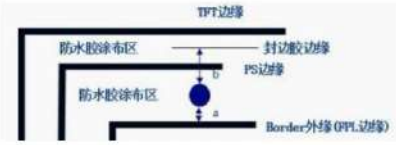
## 13. 5 General inspection standards for products

### 13.5.1 Appearance inspection standard

| Inspection item | | Figure | | A zone inspection standard | B/C zone | Inspection method | MAJ/ MIN |
|---|---|---|---|---|---|---|---|
| Spot defects | Spot defects such as dot, foreign matter, air bubble, and dent etc. | Diameter D=(L+W)/2 (L-length、W-width) Measuring method shown in the figure below   | The distance between the two spots should not be less than 10mm | 7.5"-13.3"Module（Not include 7.5"）: D>1mm N=0 0.5<D≤0.8 N≤4 D≤0.5 Ignore 0.8<D≤1 N≤2<br>4.2"-7.5"Module（Not include 4.2"）: D>0.5 N=0 0.4<D≤0.5 N≤2 D≤0.25 Ignore 0.25<D≤0.4 N≤4<br>Module below 4.2": D>0.5 N=0 0.4<D≤0.5 N≤1 D≤0.25 Ignore 0.25<D≤0.4 N≤4 0.1mm<D≤0.25 N≤3/cm² | Foreign matter D≤1mm Pass | Check by eyes Film gauge | MIN |

| Inspection item | | Figure | | A zone inspection standard | B/C zone | Inspection method | MAJ/ MIN |
|---|---|---|---|---|---|---|---|
| Line defects | Line defects such as scratch, hair etc. | L-Length, W-Width, (W/L)<1/4 Judged by line, (W/L)≥1/4 Judged by dot    | The distance between the two lines should not be less than 5mm | 7.5"-13.3"Module（Not include 7.5"）: L>10mm,N=0 W>0.8mm, N=0 5mm≤L≤10mm, 0.5mm≤W≤0.8mm N≤2 L≤5mm, W≤0.5mm Ignore<br>4.2"-7.5"Module（Not include 4.2"）: L>8mm,N=0 W>0.2mm, N=0 2mm≤L≤8mm, 0.1mm≤W≤0.2mm N≤4 L≤2mm, W≤0.1mm Ignore<br>Module below 4.2": L>5mm,N=0 W>0.2mm, N=0 2mm≤L≤5mm, 0.1mm≤W≤0.2mm N≤4 L≤2mm, W≤0.1mm Ignore | Ignore | Check by eyes Film gauge | MIN |

| Inspection item | | Figure | Inspection standard | Inspection method | MAJ/ MIN |
|---|---|---|---|---|---|
| Panel chipping and crack defects | TFT panel chipping | X the length, Y the width, Z the chipping height, T the thickness of the panel   | Chipping at the edge: Module over 7.5"（Include 7.5"）: X≤6mm,Y≤1mm Z≤T N=3 Allowed<br>Module below 7.5"(Not include 7.5"): X≤3mm,Y≤1mm Z≤T N=3 Allowed<br>Chipping on the corner: IC sideX≤2mm Y≤2mm, Non-IC sideX≤1mm Y≤1mm . Allowed<br>Note: Chipping should not damage the edge wiring. If it does not affect the display, allowed | Check by eyes、 Film gauge | MIN |
| | Crack |   | Crack at any zone of glass , Not allowed | Check by eyes、 Film gauge | MIN |
| | Burr edge |  | No exceed the positive and negative deviation of the outline dimensions X+Y≤0.2mm Allowed | Calliper | MIN |
| | Curl of panel |   Curl height | Curl height H≤Total panel length 1% Allowed | Check by eyes | MIN |

| Inspection item | | Figure | Inspection standard | Inspection method | MAJ / MIN |
|---|---|---|---|---|---|
| PS defect | Water proof film |   | 1 .Waterproof film damage, wrinkled, open edge, not allowed<br>2. Exceeding the edge of module(according to the lamination drawing) Not allowed<br>3. Edge warped exceeds height of technical file, not allowed | Check by eyes | MIN |
| RTV defect | Adhesive effect |  | Adhesive height exceeds the display surface, not allowed | Check by eyes | MIN |
| | | | 1 .Overflow, exceeds the panel side edge, affecting the size, not allowed<br>2 .No adhesive at panel edge≤1mm, mo exposure of wiring, allowed<br>3. No adhesive at edge and corner1*1mm, no exposure of wiring, allowed | | |
| | | | Protection adhesive, coverage width within W≤1.5mm, no break of adhesive, allowed | | |
| | Adhesive re-fill | | Dispensing is uniform, without obvious concave and breaking, bubbling and swell, not higher than the upper surface of the PS, and the diameter of the adhesive re-filling is not more than 8mm, allowed | Check by eyes | MIN |
| EC defect | Adhesive bubble |   | 1、 Effective edge sealing area of hot melt products ≥1/2 edge sealing area;<br>2、Bubble a+b≥1/2 effective width, N≤3, spacing≥5mm , allowed<br>No exposure of wiring, allowed | Check by eyes | MIN |

| Inspection item | | Figure | Inspection standard | Inspection method | MAJ/ MIN |
|---|---|---|---|---|---|
| EC defect | Adhesive effect |   | 1 .Overflow, exceeds the panel side edge, affecting the size, not allowed<br>2.No adhesive at panel edge≤1mm, mo exposure of wiring, allowed<br>3.No adhesive at edge and corner 1*1mm, no exposure of wiring, allowed<br>4. Adhesive height exceeds the display surface, not allowed | Visual, caliper | MIN |
| Silver dot adhesive defect | Silver dot adhesive | | 1、Single silver dot dispensing amount ≥1mm, allowed<br>2、One of the double silver dot dispensing amount is ≥1mm and the other has adhesive (no reference to 1mm) Allowed | Visual | MIN |
| | | | Silver dot dispensing residue on the panel ≤0.2mm, allowed | Film gauge | MIN |
| FPC defect | FPC wiring<br>FPC golden finger |   | FPC, TCP damage / gold finger peroxidation, adhesive residue, not allowed | Visual | MIJ |
| | | | The height of burr edge of TCP punching surface ≦ 0.4mm, not allowed | Caliper | MIN |
| | FPC damage/crease |  | Damage and breaking, not allowed<br>Crease does not affect the electrical performance display, allowed | Check by eyes | MIN |

| Inspection item | | Figure | Inspection standard | Inspection method | MAJ/ MIN |
|---|---|---|---|---|---|
| Protective film defect | Protective film | Scratch and crease on the surface but no affect to protection function, allowed | | Check by eyes | MIN |
| | | Adhesive at edge L≤5mm， W≤0.5mm， N=2, no entering into viewing area | | Check by eyes | MIN |
| Stain defect | Stain | If stain can be normally wiped clean by > 99% alcohol, allowed | | Visual | MIN |
| Pull tab defect | Pull tab | The position and direction meet the document requirements, and ensure that the protective film can be pulled off. | | Check by eyes/ Manual pulling | MIN |
| Shading tape defect | Shading tape | Tilt≤10°, flat without warping, completely covering the IC. | | Check by eyes/ Film gauge | MIN |
| Stiffener | Stiffener | Flat without warping, Exceeding the left and right edges of the FPC is not allowed. Left and right can be less than 0.5mm from FPC edge | | Check by eyes | MIN |
| Label | Label/ Spraying code | The content meets the requirements of the work sheet. The attaching position meets the requirements of the technical documents. | | Check by eyes | MIN |

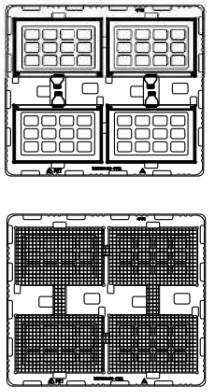
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## 14. Packing

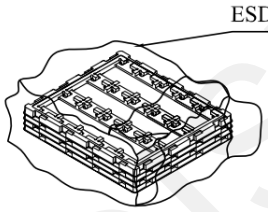
| CUSTOMER'S APPROVED: | | PAGE: 1/1 |
|---|---|---|

PACKLING ORDER:

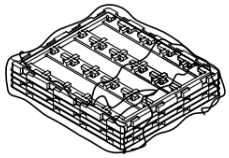
1) Putting 4 pcs Modules on each PET tray.And cover a dedicated EPE film.



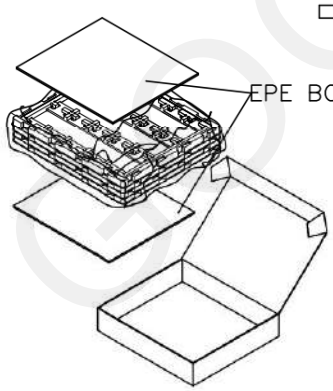
2) Putting 10 pcs PET trays together with 1 empty tray on the top of PET tray. Insert in the ESD bag, add desiccant in the ESD bag.



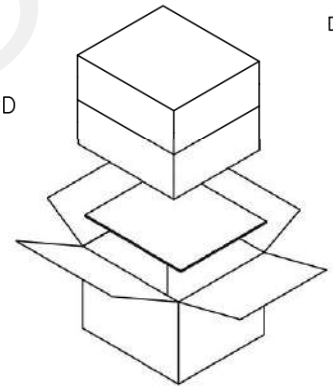

3) the tray together with rubber band



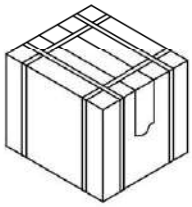
4) Putting in the inner small carton




5) Putting 2 small cartons into one outcarton



6) Packing finished



Note:4 pcs in a tray, 10 trays in a inner carton, 2 inner cartons in a out carton, so 4x10x2=80pcs/Outcarton

Dimension (Small carton ): 410*353*112mm  Dimension (Out carton ): 424*374*250mm

| | Drw: | | Chk: | | Apv: | |
|---|---|---|---|---|---|---|

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## 15. Precautions

(1) Do not apply pressure to the EPD panel in order to prevent damaging it.

(2) Do not connect or disconnect the interface connector while the EPD panel is in operation.

(3) Do not touch IC bonding area. It may scratch TFT lead or damage IC function.

(4) Please be mindful of moisture to avoid its penetration into the EPD panel, which may cause damage during operation.

(5) High temperature, high humidity, sunlight or fluorescent light may degrade the EPD panel's performance. Please do not expose the unprotected EPD panel to high temperature, high humidity, sunlight, or fluorescent for long periods of time.

(6) For more precautions, please click on the link:
*https://www.good-display.com/news/80.html*

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